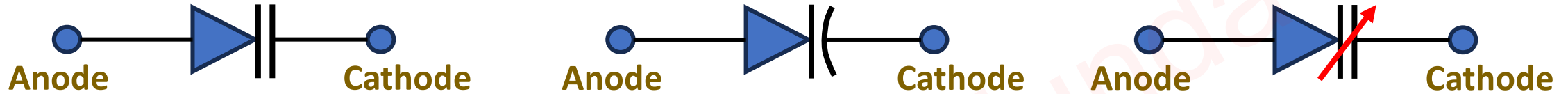


Varactor Diode

Symbols of Varactor Diode



Basics of Varactor Diode

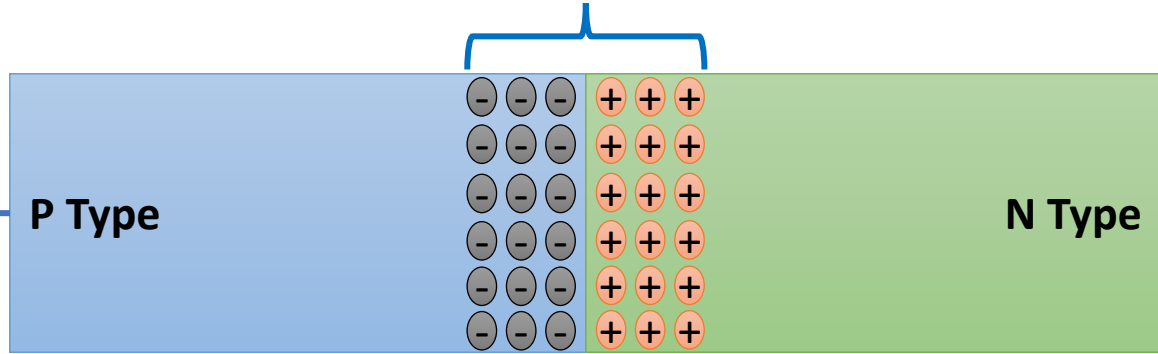
- ❑ Varactor Diode is also known as Varicap Diode, Variable Capacitance Diode, or Tuning Diode.
- ❑ Varactor Diode was developed in 1961 by Pacific Semiconductor.
- ❑ Varactor Diode works in the reverse bias.
- ❑ In the reverse bias, depletion width Increases which Decreases the capacitance as per:

$$C = \frac{\epsilon_0 \epsilon_r A}{W}$$

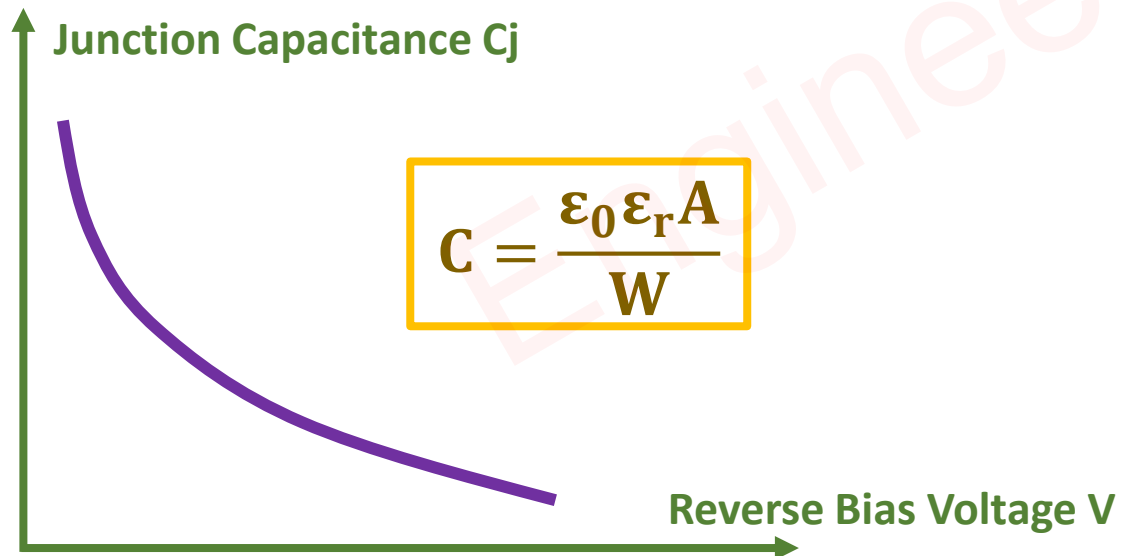
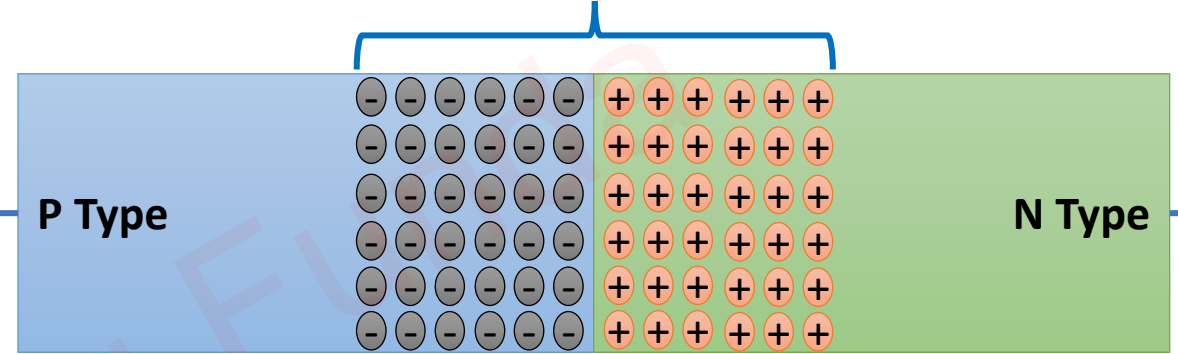
- ❑ Varicap means Variable Capacitance and Varactor means Variable Reactance.

Working & Characteristics of Varactor Diode

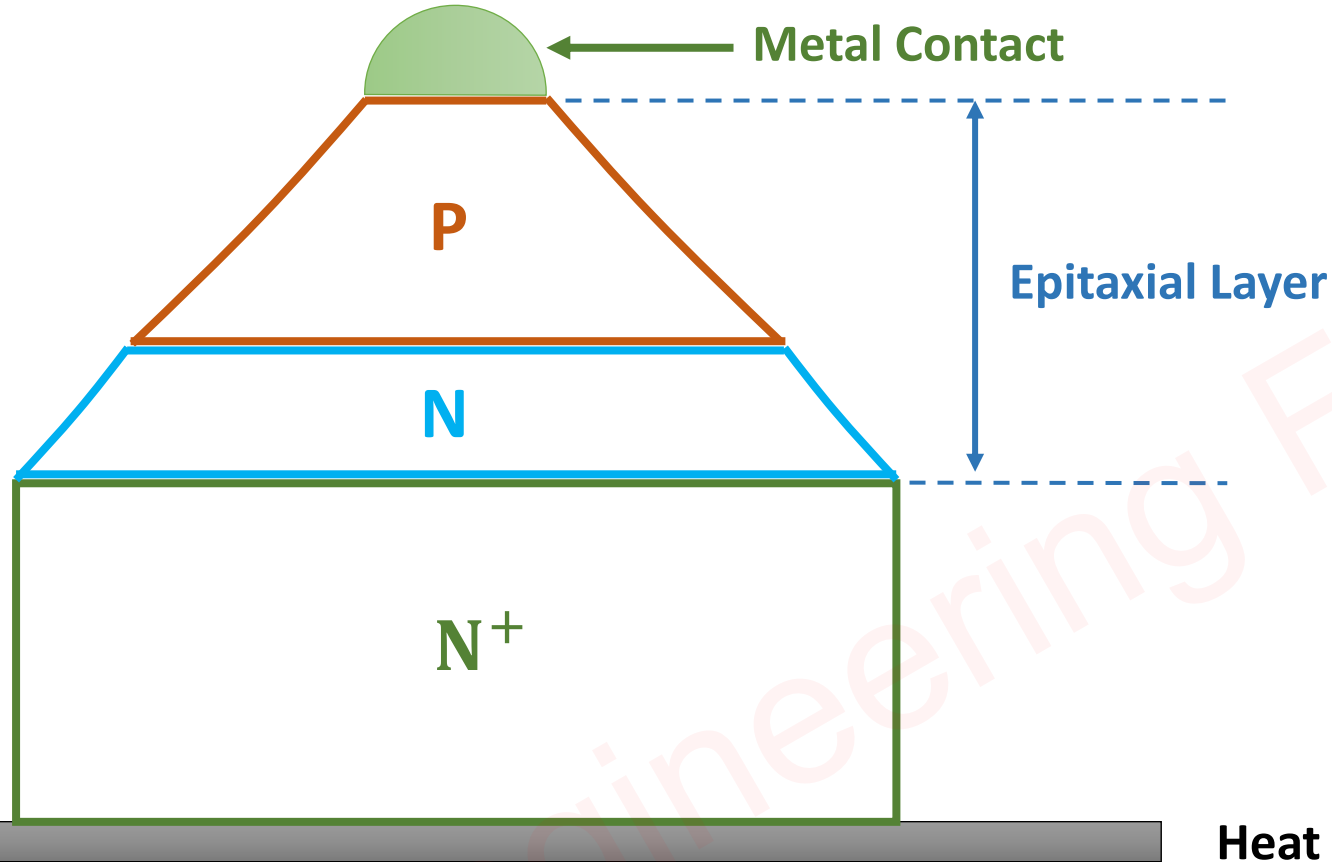
Space Charge region or
depletion region W



Space Charge region or
depletion region W



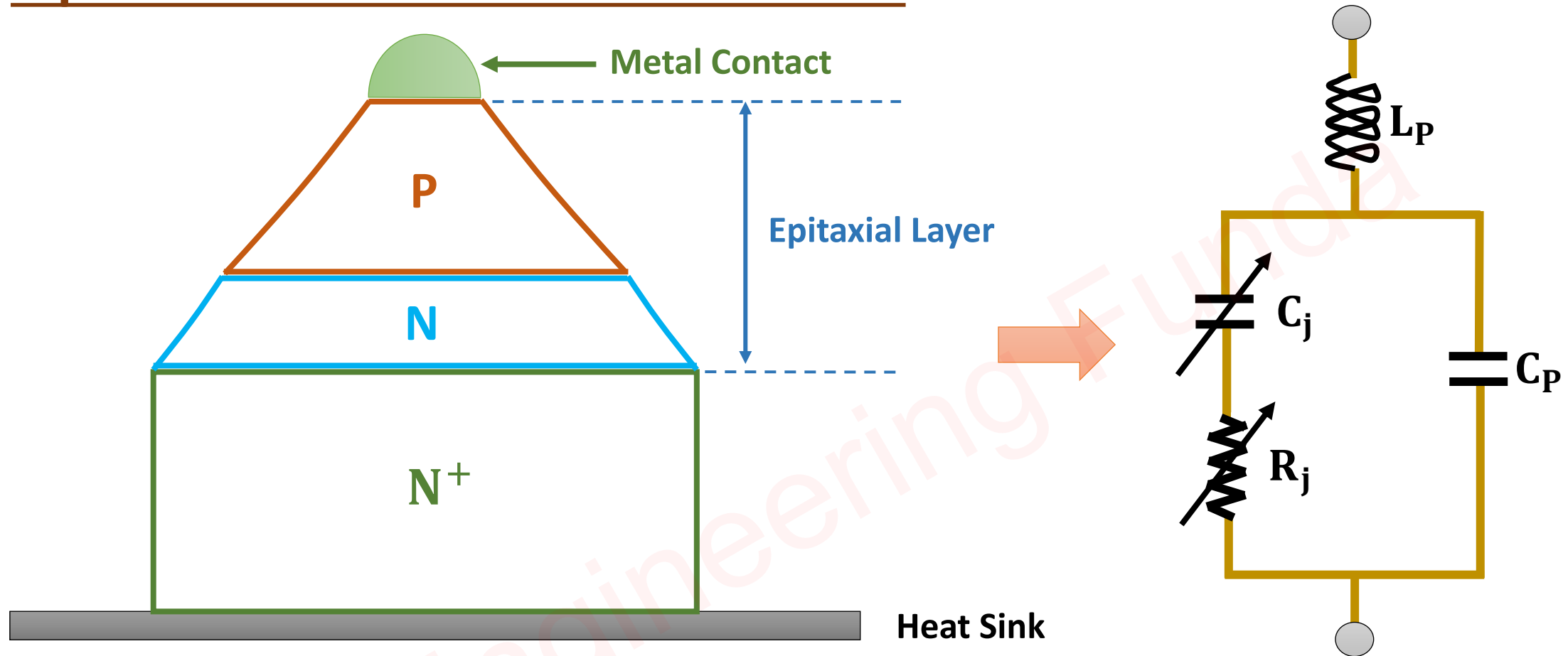
Structure of Varactor Diode



$$C = \frac{\epsilon_0 \epsilon_r A}{W}$$

- ☐ To have a Linear relation of Reverse Bias Voltage and Junction Capacitance Epitaxial Layer is provided.
- ☐ N⁺ Region is for the Drifting of Electrons.
- ☐ Due to the N⁺ Region, the Varactor Diode can function at Microwave Frequencies.

Equivalent Circuit of Varactor Diode



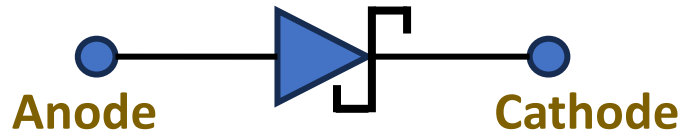
- ❑ Junction Capacitance C_j and Junction Resistance R_j changes based on Reverse Bias voltage.
- ❑ Cap Capacitance C_P and Cap Inductance L_P is present due to metallic contacts of Varactor Diode.

Applications of Varactor Diode

- ❑ Parametric Amplifiers
- ❑ Frequency Multiplier
- ❑ Tuning Circuits
- ❑ Voltage Controlled Oscillator
- ❑ RF Phase Shifter
- ❑ Frequency and Phase Modulator
- ❑ Adjustable Bandpass Filter
- ❑ Frequency Synthesizers

Schottky Diode

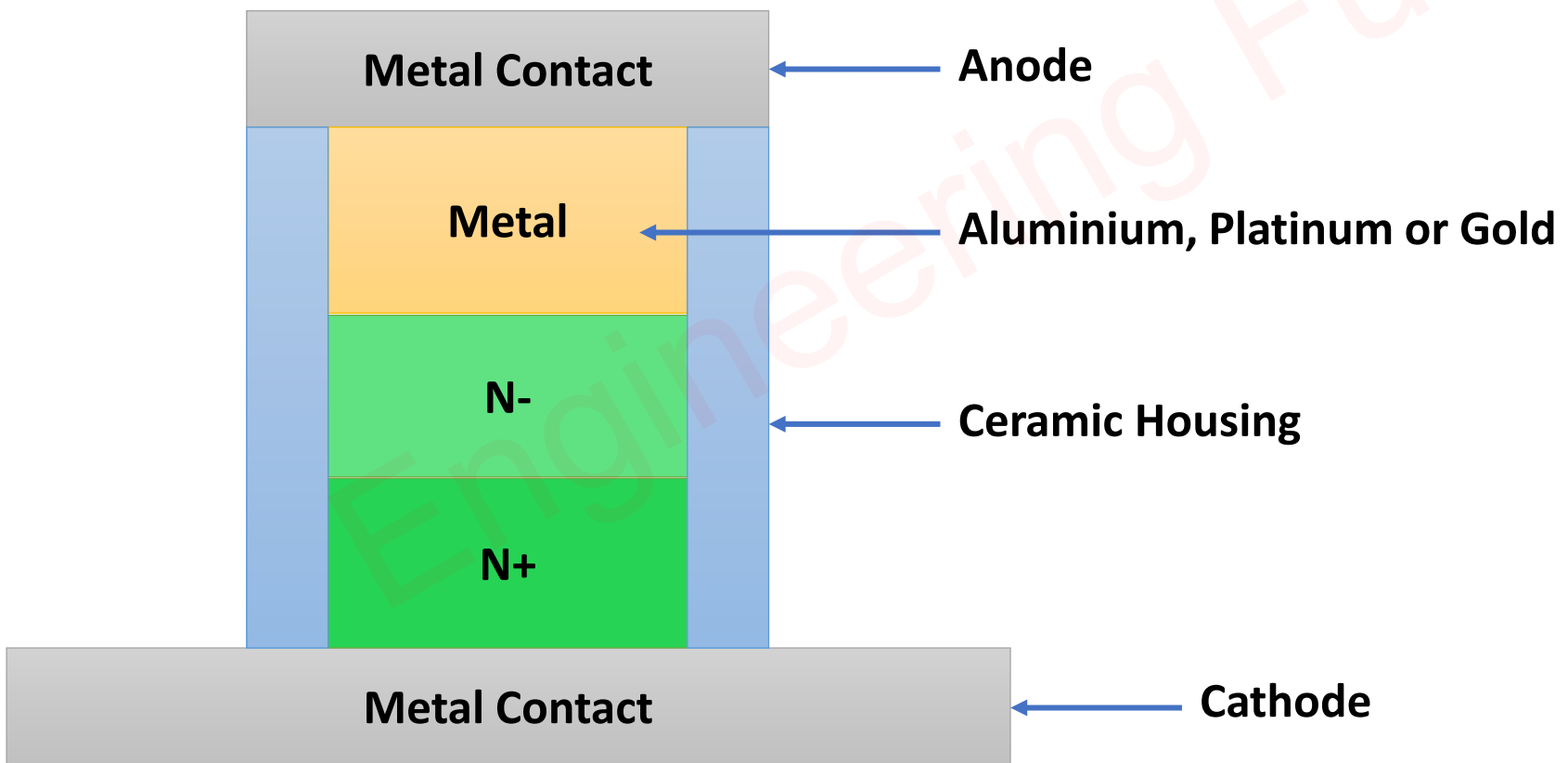
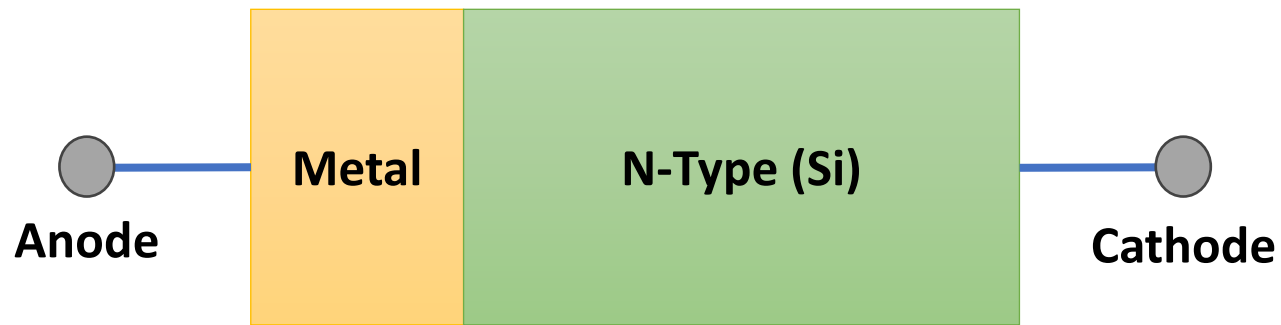
Symbol of Schottky Diode



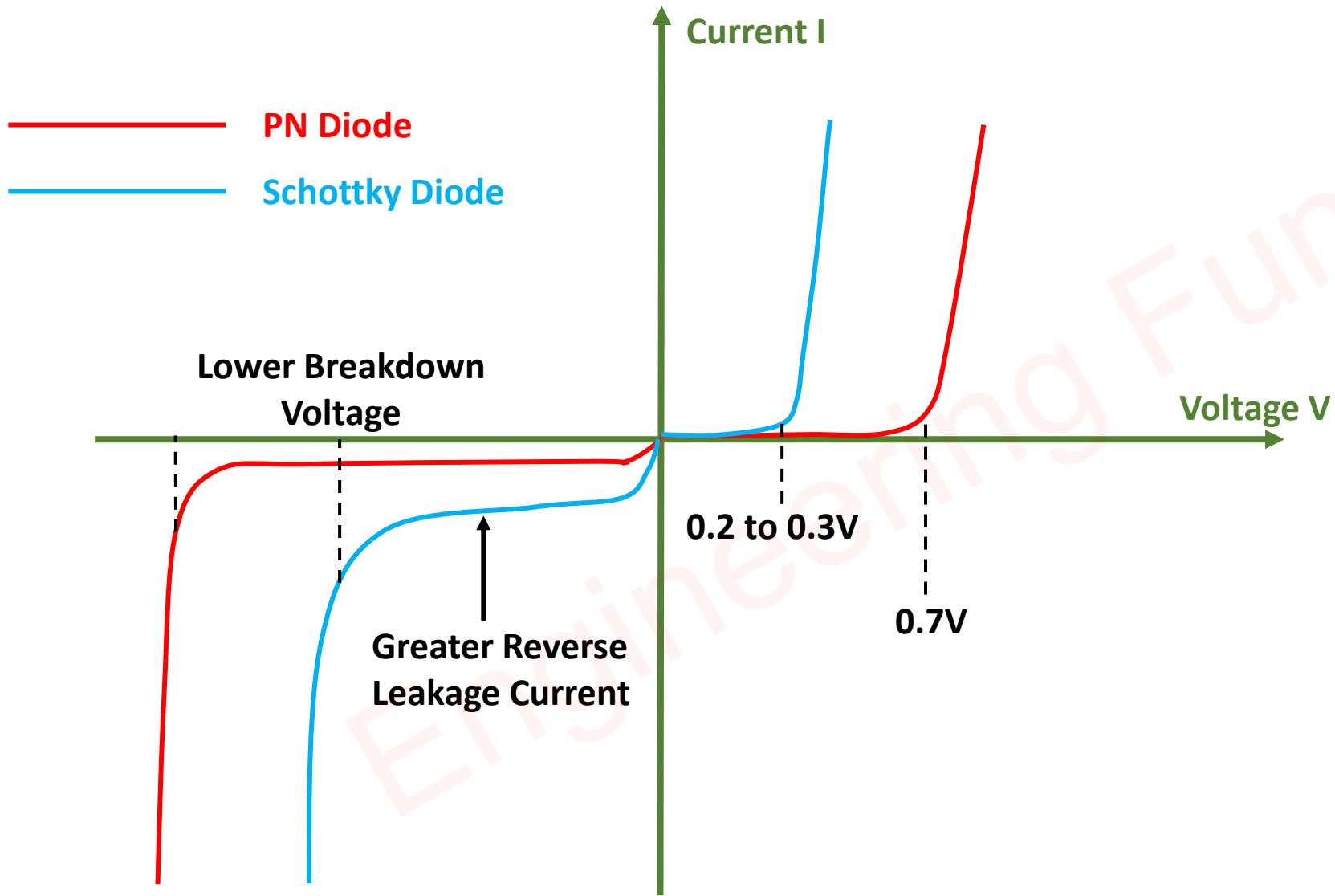
Basics of Schottky Diode

- ☐ Schottky Diode is also known as Schottky Barrier Diode.
- ☐ Unlike standard diodes, with a p-n junction, a Schottky diode has a metal-semiconductor junction.
- ☐ The metal is typically aluminum, platinum, or gold, while the semiconductor is usually n-type silicon. Hence, the Majority of Charge Carriers are electrons Only.
- ☐ Schottky Diode has fast switching characteristics compared to Standard PN Junction Diode, Hence It is useful in RF/Microwave Circuits.
- ☐ Schottky Diode has a low Forward Voltage Drop (0.2V to 0.3V) compared to Silicon PN Junction (0.7V) Diode.

Structure of Schottky Diode



Characteristics of Schottky Diode



Features of Schottky Diode

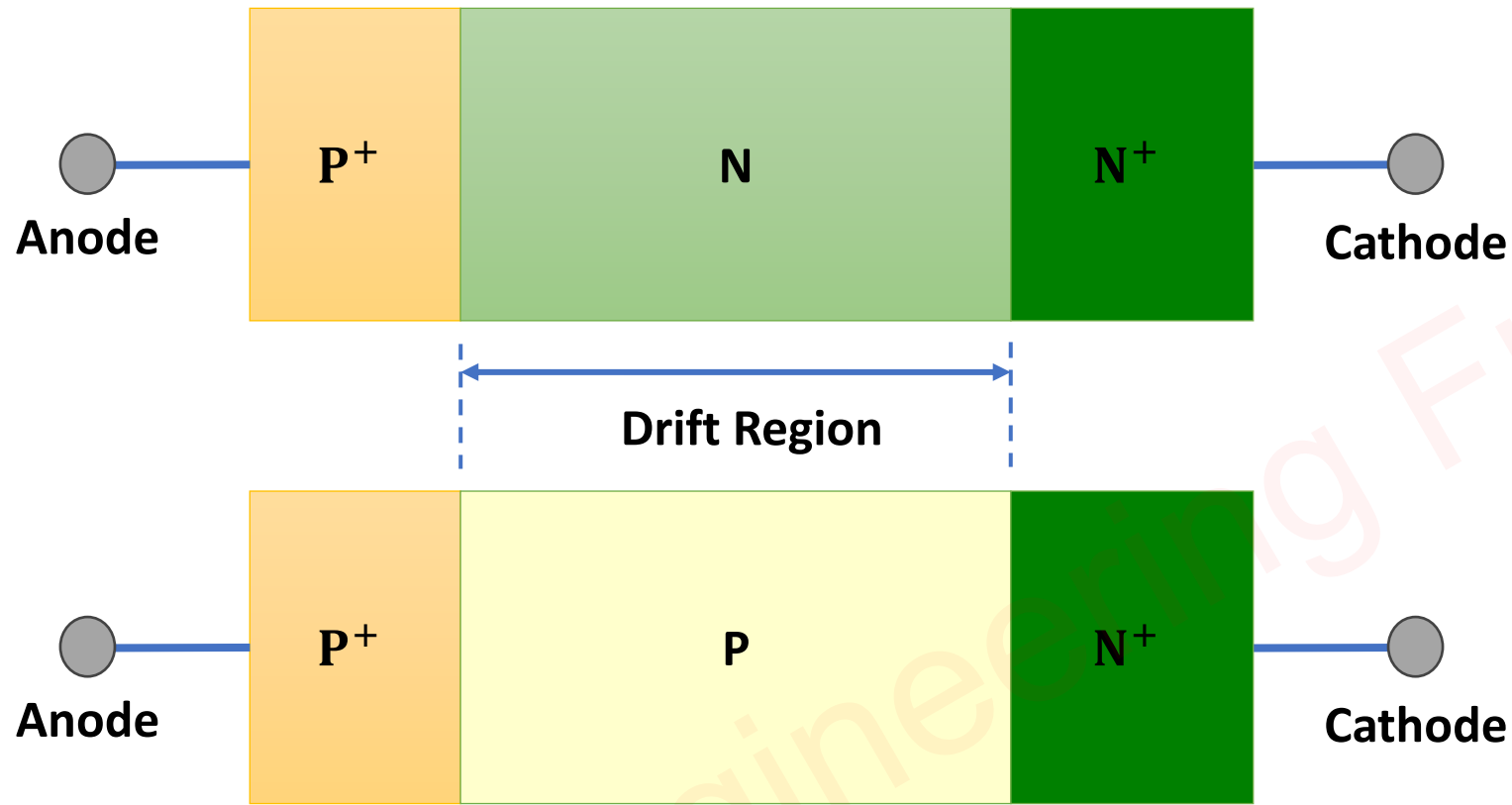
- ❑ **Low Forward Voltage Drop:** Schottky diodes have a lower forward voltage drop (typically 0.2 to 0.3 volts) compared to silicon diodes (around 0.7 volts). This makes them more efficient for applications requiring fast switching and low power loss.
- ❑ **Fast Switching Speed:** The absence of minority carrier charge storage leads to faster switching speeds, making Schottky diodes suitable for high-frequency applications.
- ❑ **Low Reverse Recovery Time:** Schottky diodes have negligible reverse recovery time, which is the time it takes for the diode to switch from conducting to blocking mode.
- ❑ **More Reverse Leakage Current and Less PIV:** Schottky diode has more Reverse Leakage Current and Less PIV, Hence Schottky Diodes are suitable for low-power and Fast switching applications.

TRAPATT Diode

Basics of TRAPATT Diode

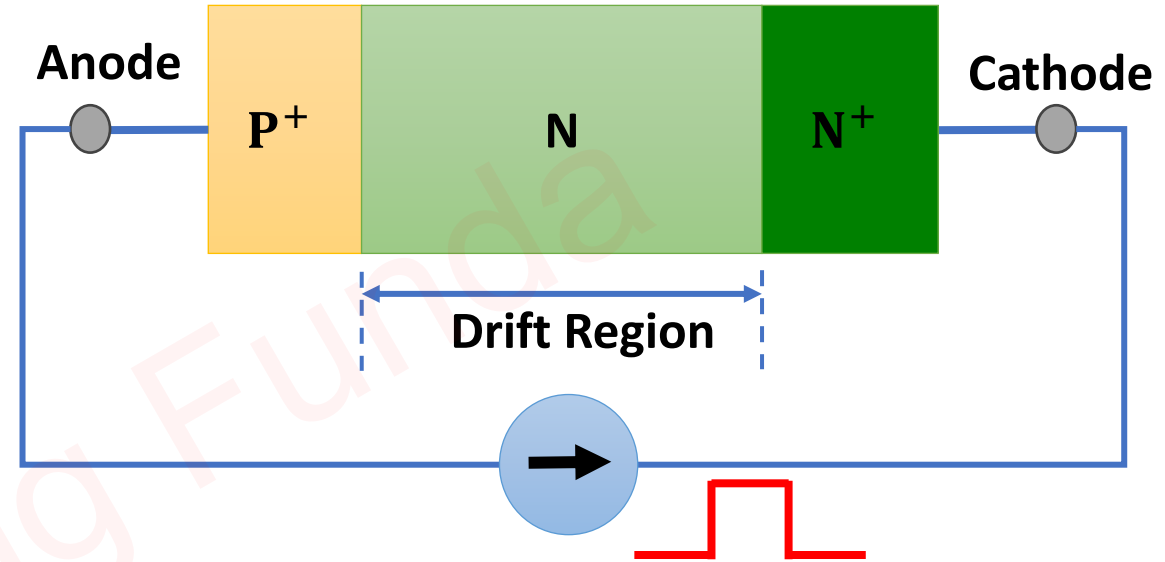
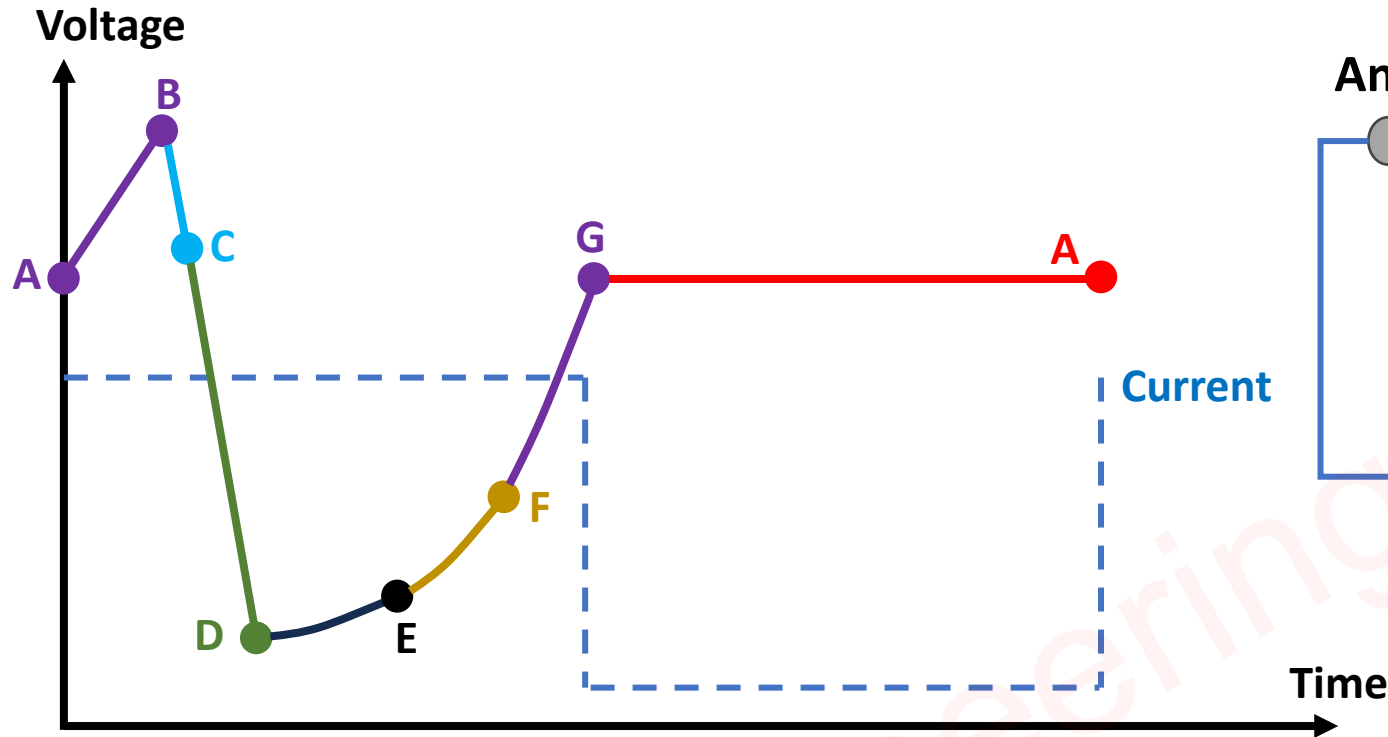
- ❑ TRAPATT Diode is **TR**apped **P**lasma **A**valanche **T**riggered **T**ransit Diode.
- ❑ TRAPATT Diode is operated by the creation of Plasma {High-Density Charge Carriers}.
- ❑ TRAPATT Diode is used in RF/Microwave Applications. Typically, they are used as Oscillators and Amplifiers for the Frequency range of 0.5 GHz to 10 GHz.
- ❑ TRAPATT Diode has a Noise Figure of around 60dB.
- ❑ TRAPATT Diode structure consist of $(P^+ - N - N^+)$ or $(N^+ - P - P^+)$ Semiconductor Layers.
- ❑ TRAPATT Diode gives higher efficiency than IMPATT Diode but also has a higher noise.
- ❑ In reverse bias, It causes Impact Ionization {Cause Strong E-Field due to Ions} and generates a high-density plasma, which can be used for oscillations.

Structure of TRAPATT Diode



- ☐ The Middle Layer is lightly doped.
- ☐ The Middle Layer is the Drift Region.
- ☐ Plasma is getting trapped in the Drift Region.
- ☐ Impact Ionization happens at the side of a Highly doped region.

Working of TRAPATT Diode



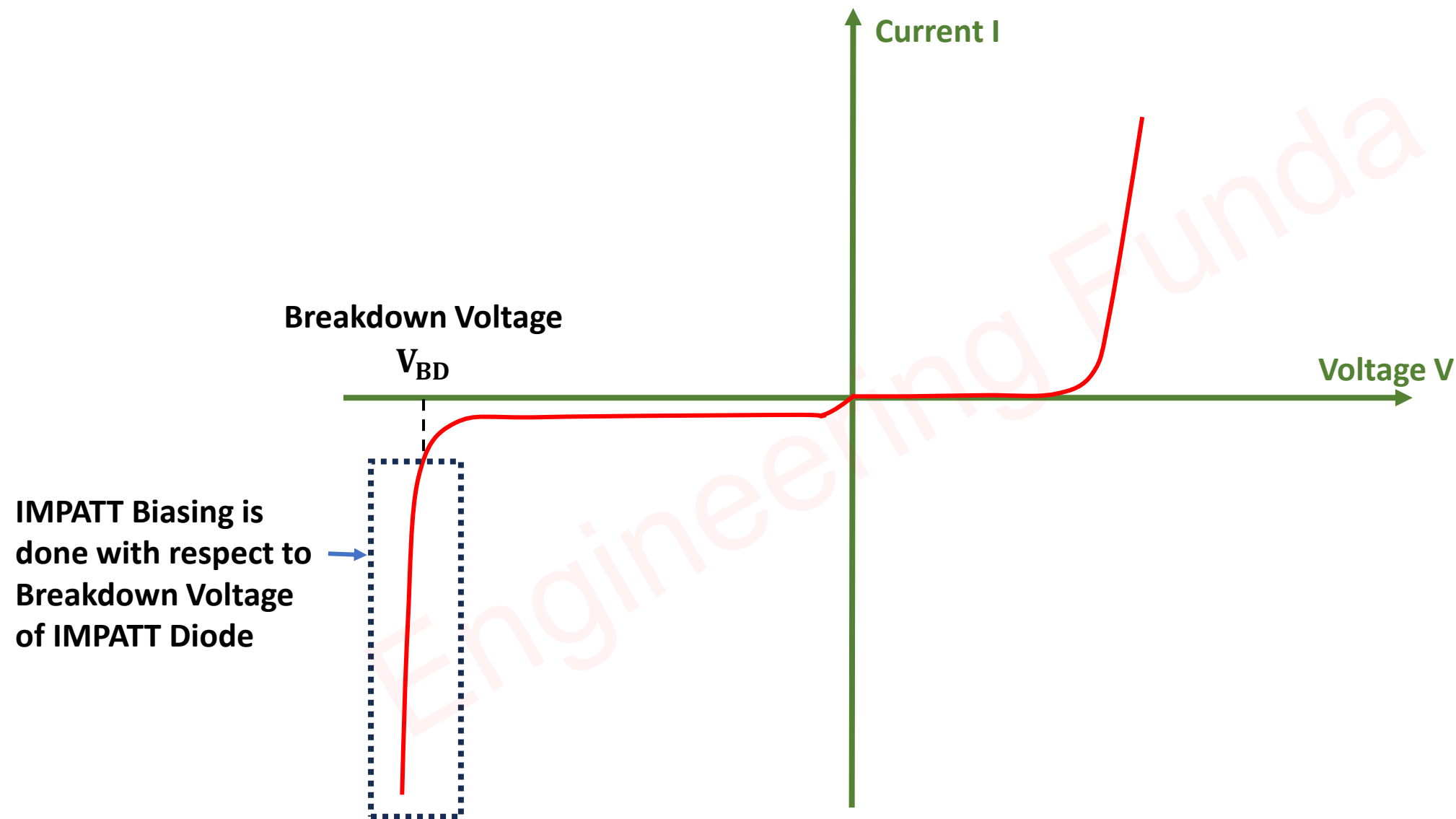
- ❑ A to B: The Diode Acts as a capacitor and it gets charge from A to B.
- ❑ B to C: Plasma Formation in Drift Region.
- ❑ C to D: Creation of Dense Charge Particles {Electrons and Holes} in Drift Region.
- ❑ D to E: Plasma Extraction.
- ❑ E to F: Residual Extraction, Dense Charge Particles comes out of the Drift Region.
- ❑ F to G: The Acts as a capacitor and it gets charge from F to G.
- ❑ In TRAPPT Diode, the current leads the voltage by approximately 90 degrees.

IMPATT Diode

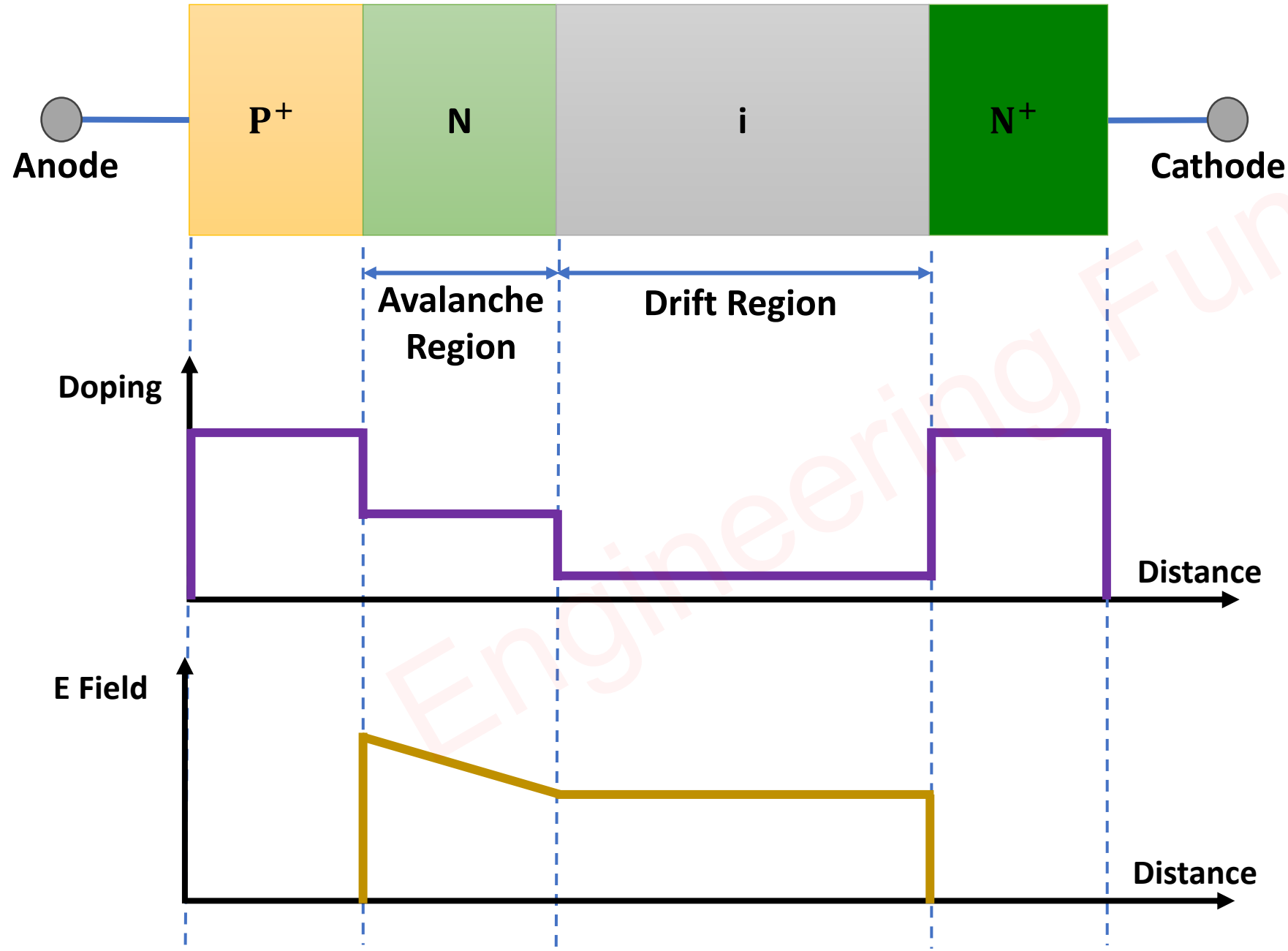
Basics of IMPATT Diode

- ❑ IMPATT Diode is **IMP**act **I**onization **A**valanche **T**ransit **T**ime Diode.
- ❑ Read proposes IMPATT Diode, So IMPATT Diode is also known as Read Diode.
- ❑ It is a High Power Microwave Diode.
- ❑ IMPATT Diode has a frequency range of 1GHz to 100GHz.
- ❑ IMPATT Diode has a High Power Handling Capacity.
- ❑ IMPATT Diode has Negative Resistance Characteristics, which can be utilized in Microwave Oscillators.
- ❑ The drawback of IMPATT Diode is Higher Phase Noise.
- ❑ IMPATT Diode is mainly used in Reverse Bias {in Avalanche Break Down Region}, as IMPATT Diode has lower reverse recovery time.

Characteristics of IMPATT Diode

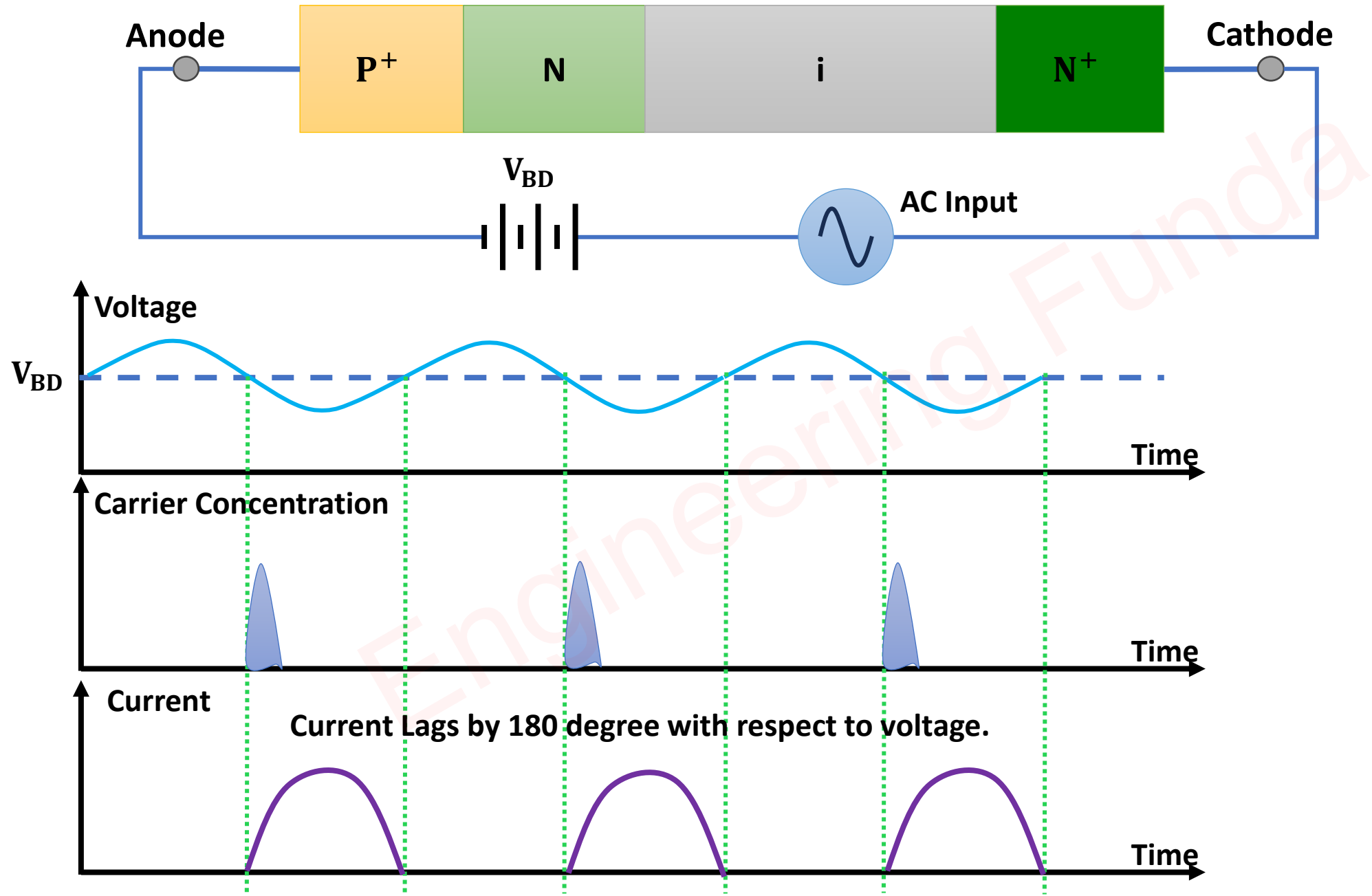


Structure of IMPATT Diode



- ❑ IMPATT Diode has vertical structure, which leads to high power handling capacity of IMPATT Diode.
- ❑ IMPATT Diode can be fabricated using GaAs, Si, InP or SiC.
- ❑ Avalanche Region has extreme high E field, where charge multiplication happens in Reverse bias breakdown.
- ❑ After Reverse Breakdown, Electrons moves towards N^+ region and Holes moves towards P^+ region.

Working of IMPATT Diode



Applications of IMPATT Diode

- ❑ Microwave Oscillators
- ❑ Microwave Amplifiers
- ❑ Frequency Multipliers
- ❑ RADAR Systems
- ❑ Satellite Communication
- ❑ Research and Education

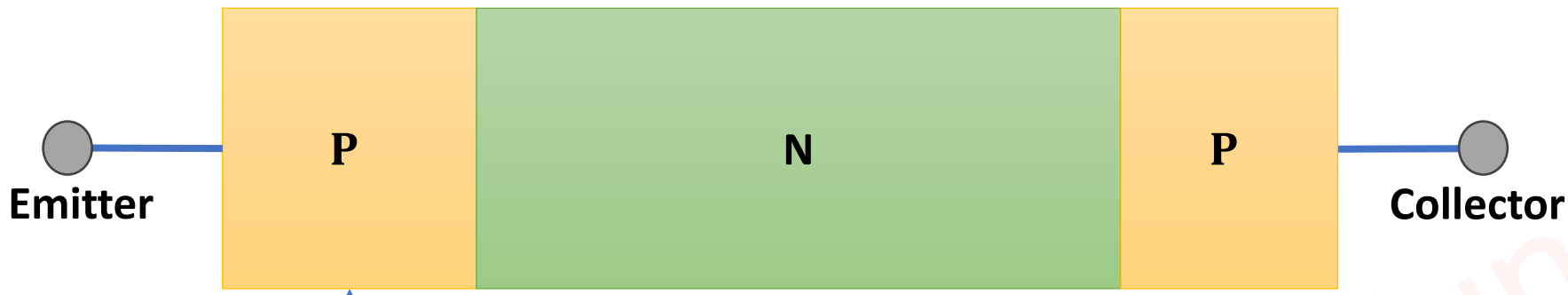
Engineering Funda

BARITT Diode

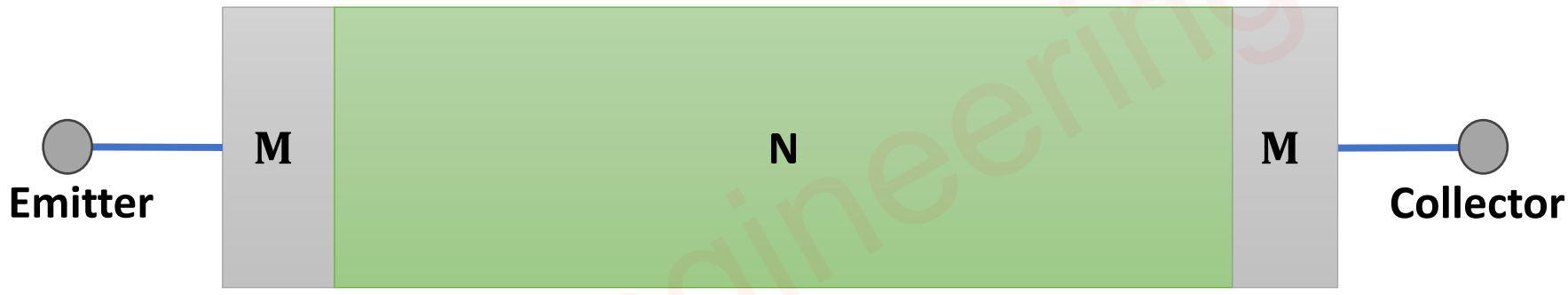
Basics of BARITT Diode

- ❑ BARITT Diode is **BAR**rier **I**njection **T**ransit **T**ime Diode.
- ❑ BARITT Diode functions based on Thermionic Emission.
- ❑ In BARITT Diode, Current flows based on Punch Through Effect.
- ❑ The BARITT Diode has Lower noise than the IMPATT Diode and TRAPATT Diode.
- ❑ BARITT Diode works at frequency Range of 4GHz to 8GHz.
- ❑ BARITT Diode available with P-N-P or M-N-M or P-N-i-P Layers.
- ❑ BARITT Diode offers Negative Resistance Characteristics.
- ❑ BARITT Diodes are available with Si and GaAs materials.

Structure of BARITT Diode

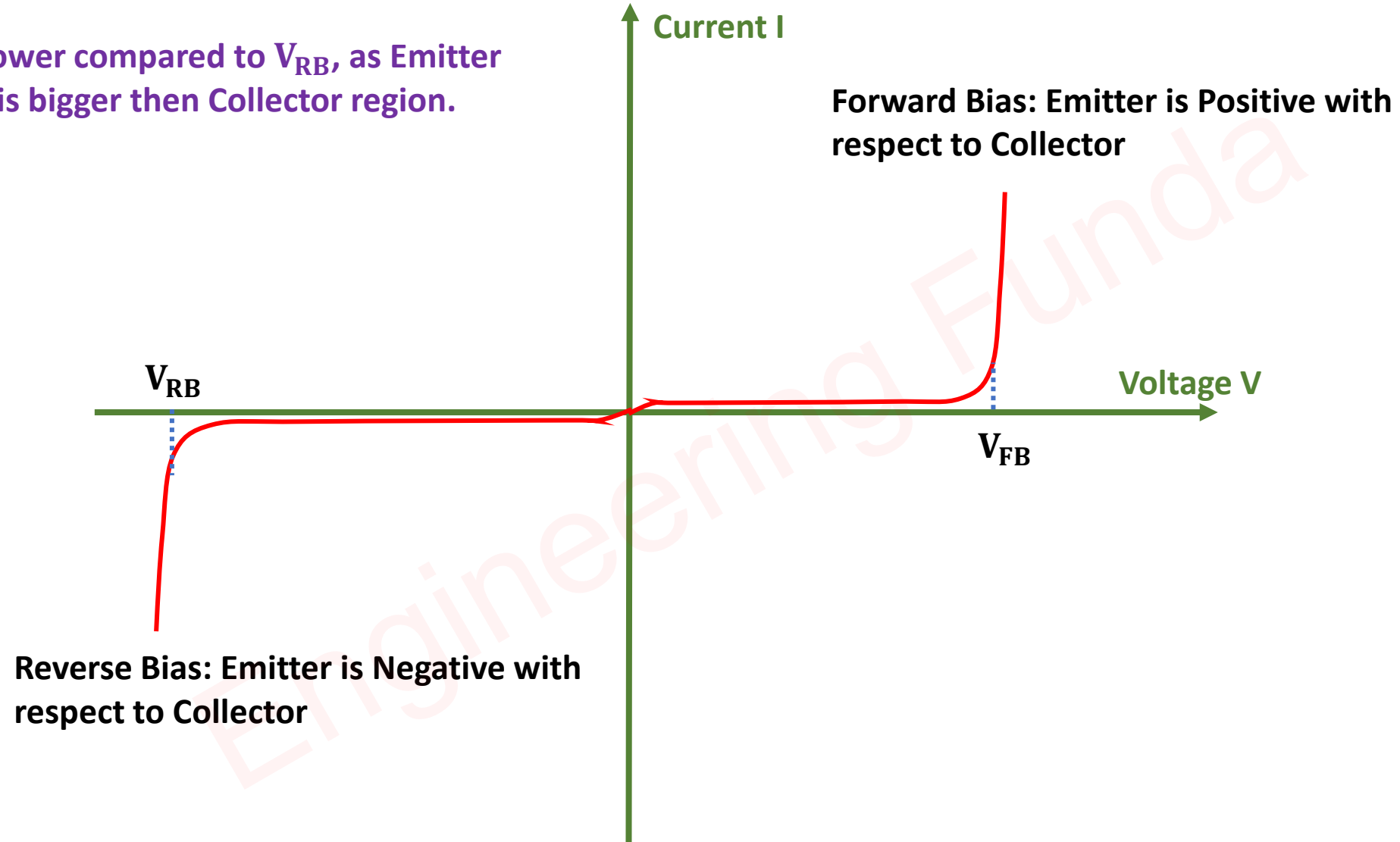


Larger Region Compared to Collector

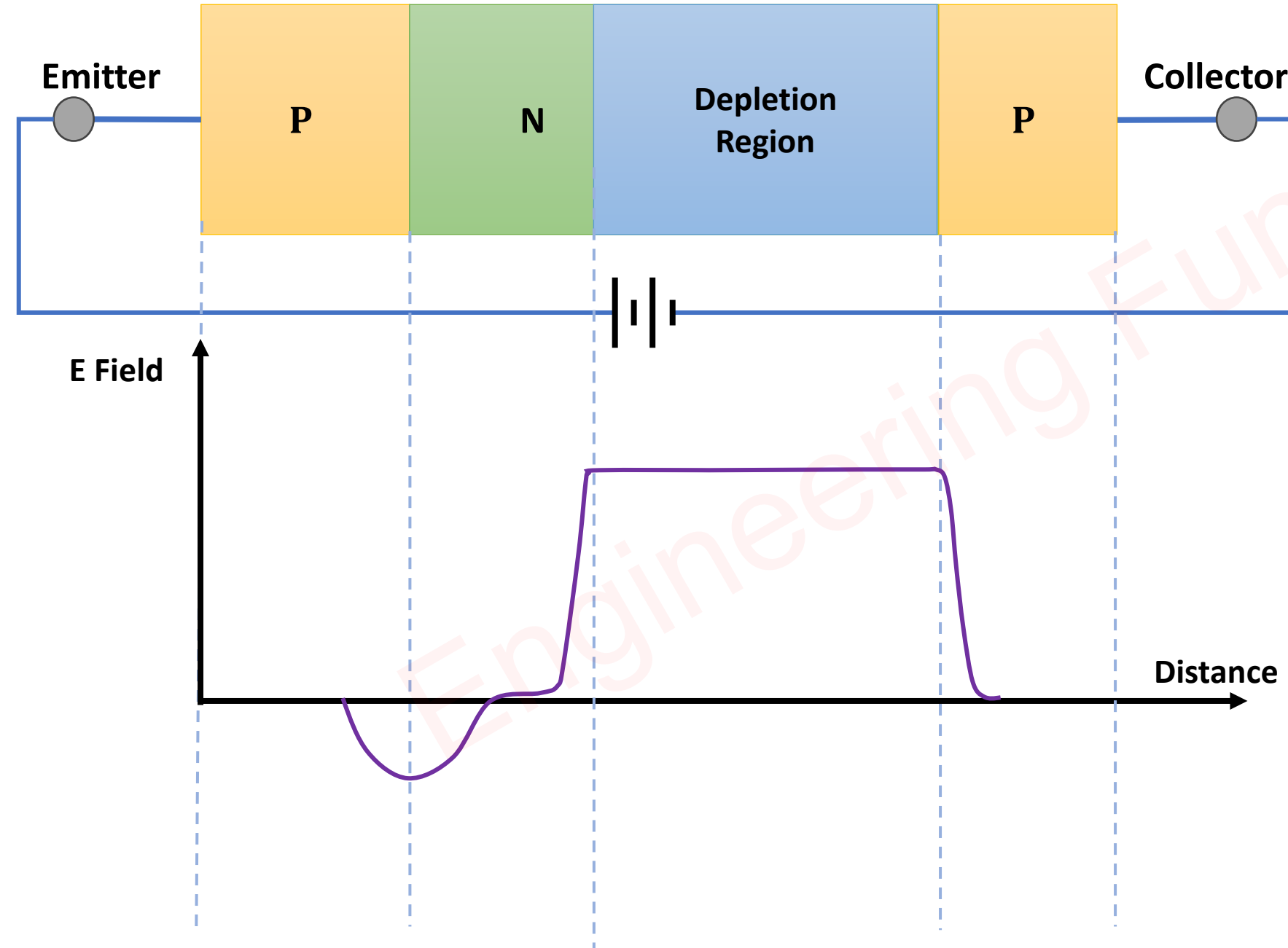


Characteristics of BARITT Diode

- V_{FB} is lower compared to V_{RB} , as Emitter Region is bigger than Collector region.

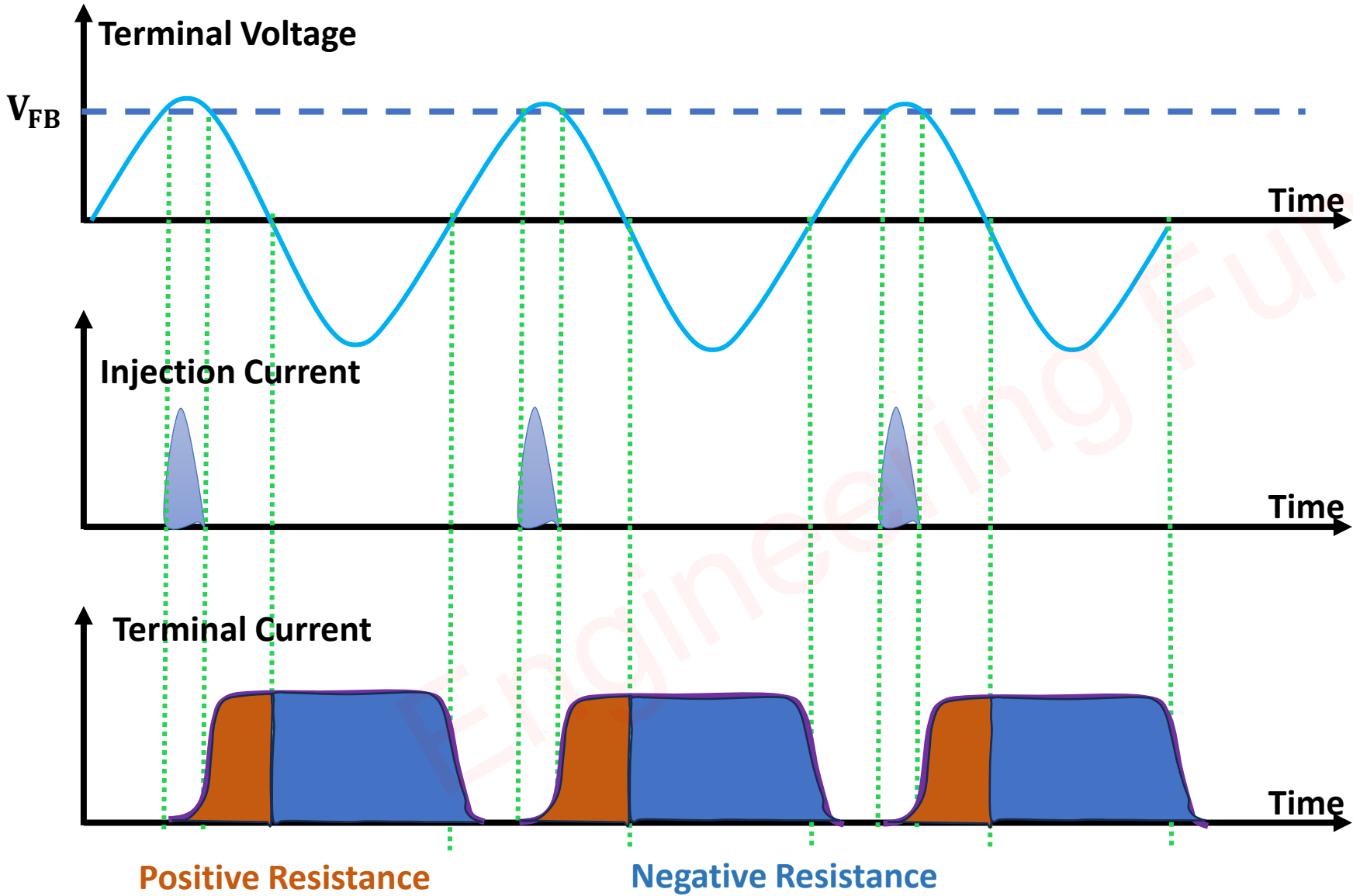


Punch Through in BARITT Diode



- ❑ If we increase the Forward bias voltage beyond some limits, then Holes of the Emitter Punch through towards the collector.
- ❑ In that case, E Field profile has no negative value at the forward bias Emitter junction.

Working of BARITT Diode



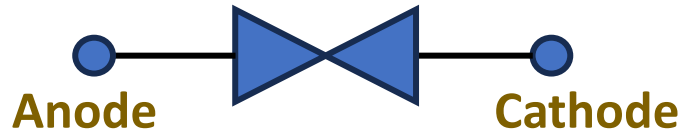
Applications of BARITT Diode

- ❑ Microwave Oscillators
- ❑ Low Noise Signal Generator
- ❑ Microwave Mixer

Engineering Funda

Gunn Diode

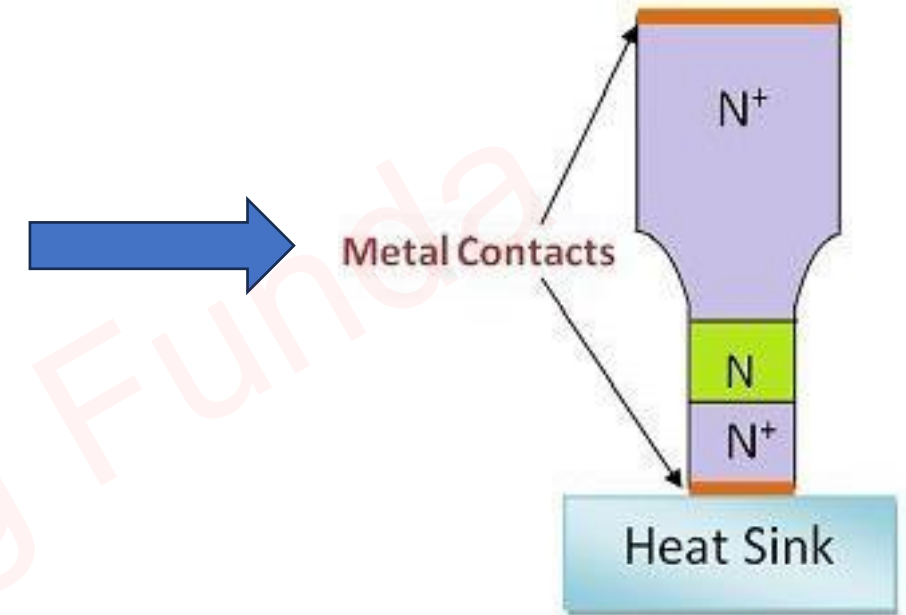
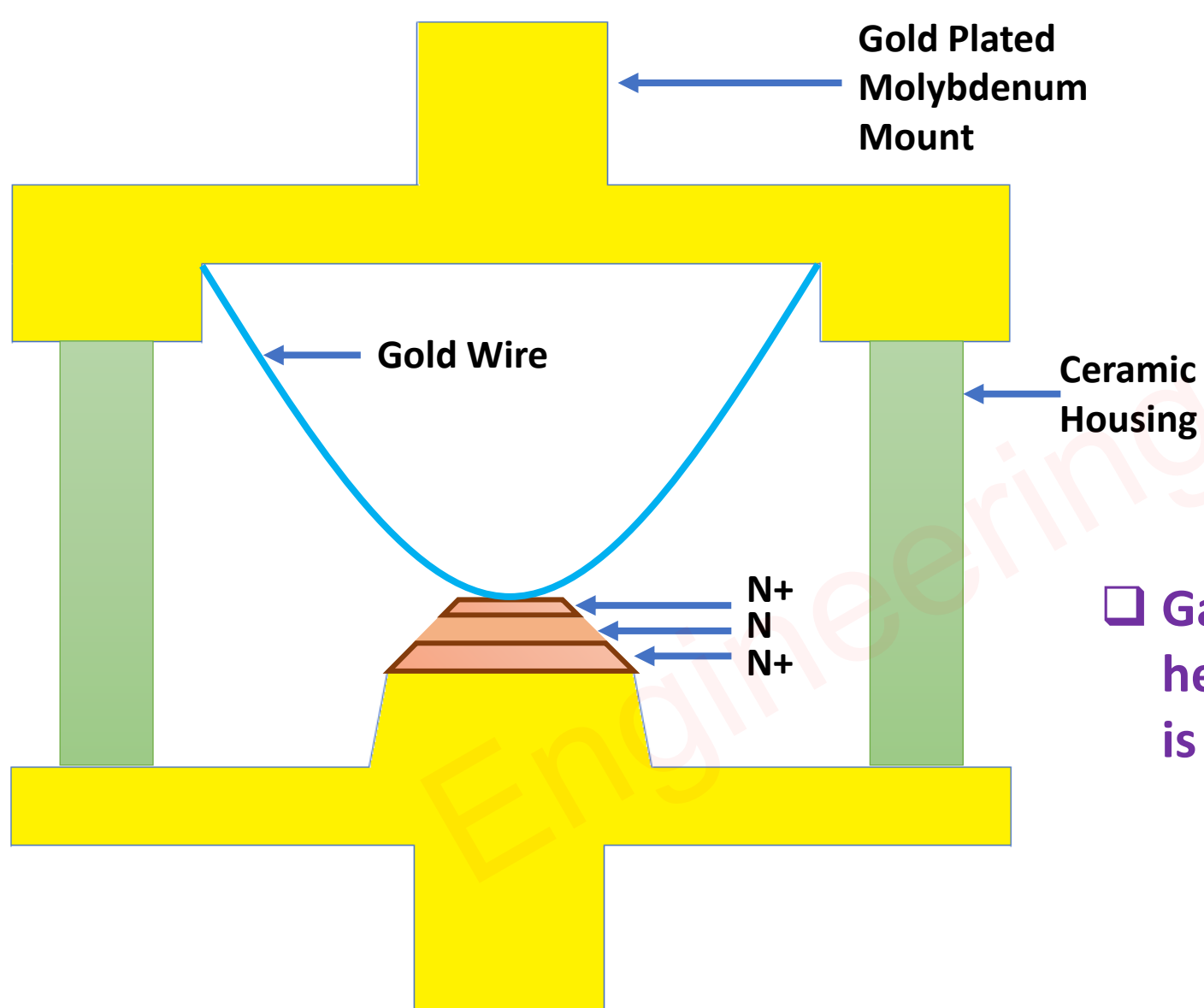
Symbol of Gunn Diode



Basics of Gunn Diode

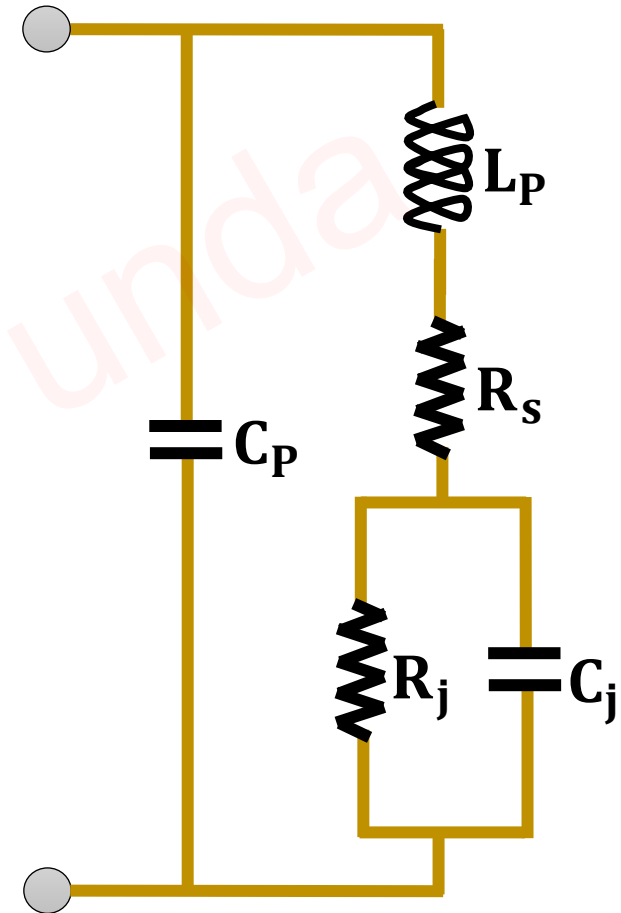
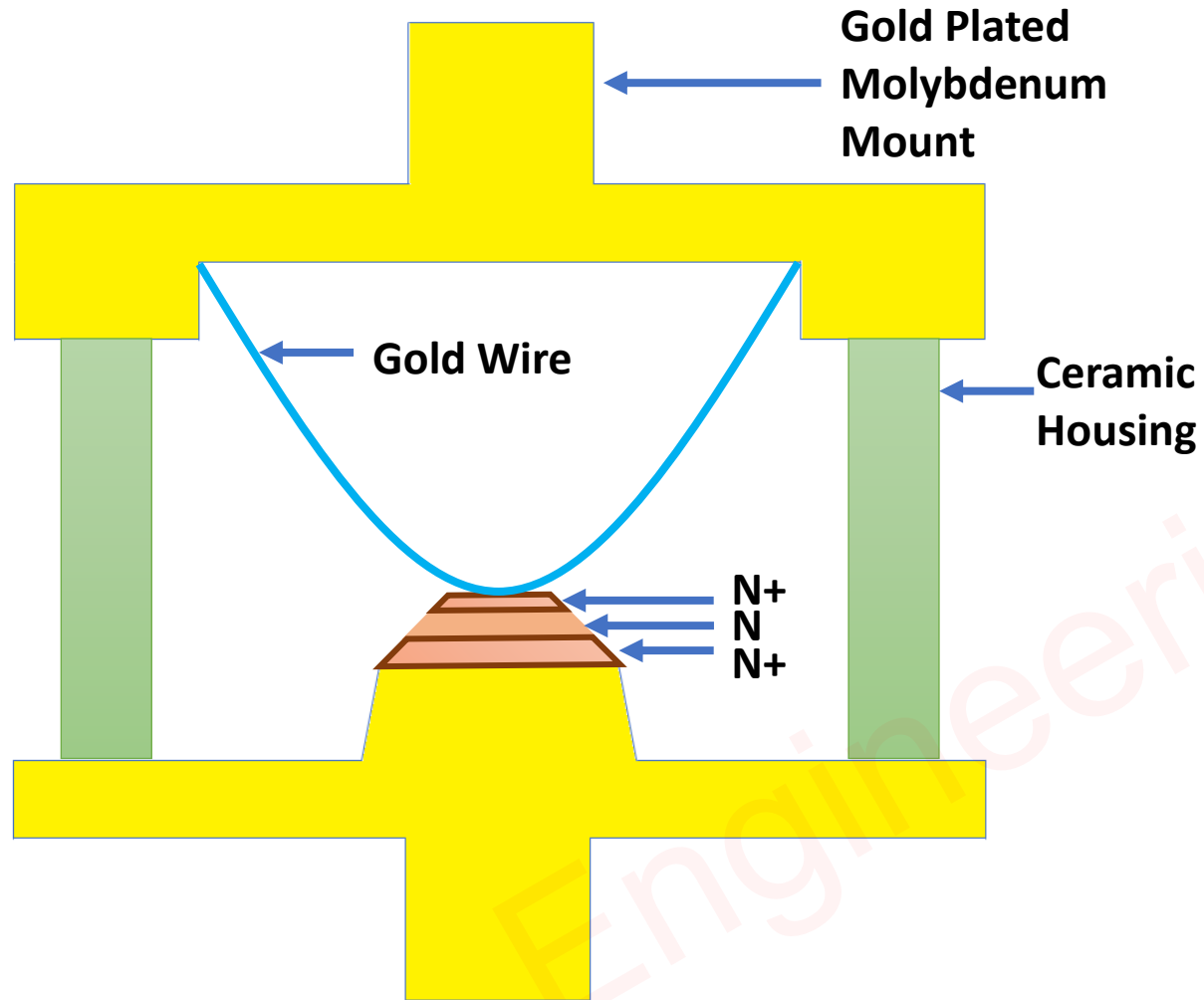
- ☐ Gunn Diode was discovered by **JB Gunn** in 1962.
- ☐ Gunn Diode can be fabricated by GaAs, InP, InSb {Indium Antimonide}, ZnSe {Zinc Selenide} Materials.
- ☐ Gunn Diode is not a PN Junction Diode, it is based on only N Type Layer, Hence it has only electrons as Majority charge carrier.
- ☐ Gunn Diode offers **Negative Resistance Characteristics**, that result in the usage of Gunn Diode at RF/Microwave signals.
- ☐ Gunn Diode works based on the “**Gunn Effect**” or “**Transferred Electron Effect**”. Hence, it is also called as “**Transferred Electron Device - TED**”.

Structure of Gunn Diode



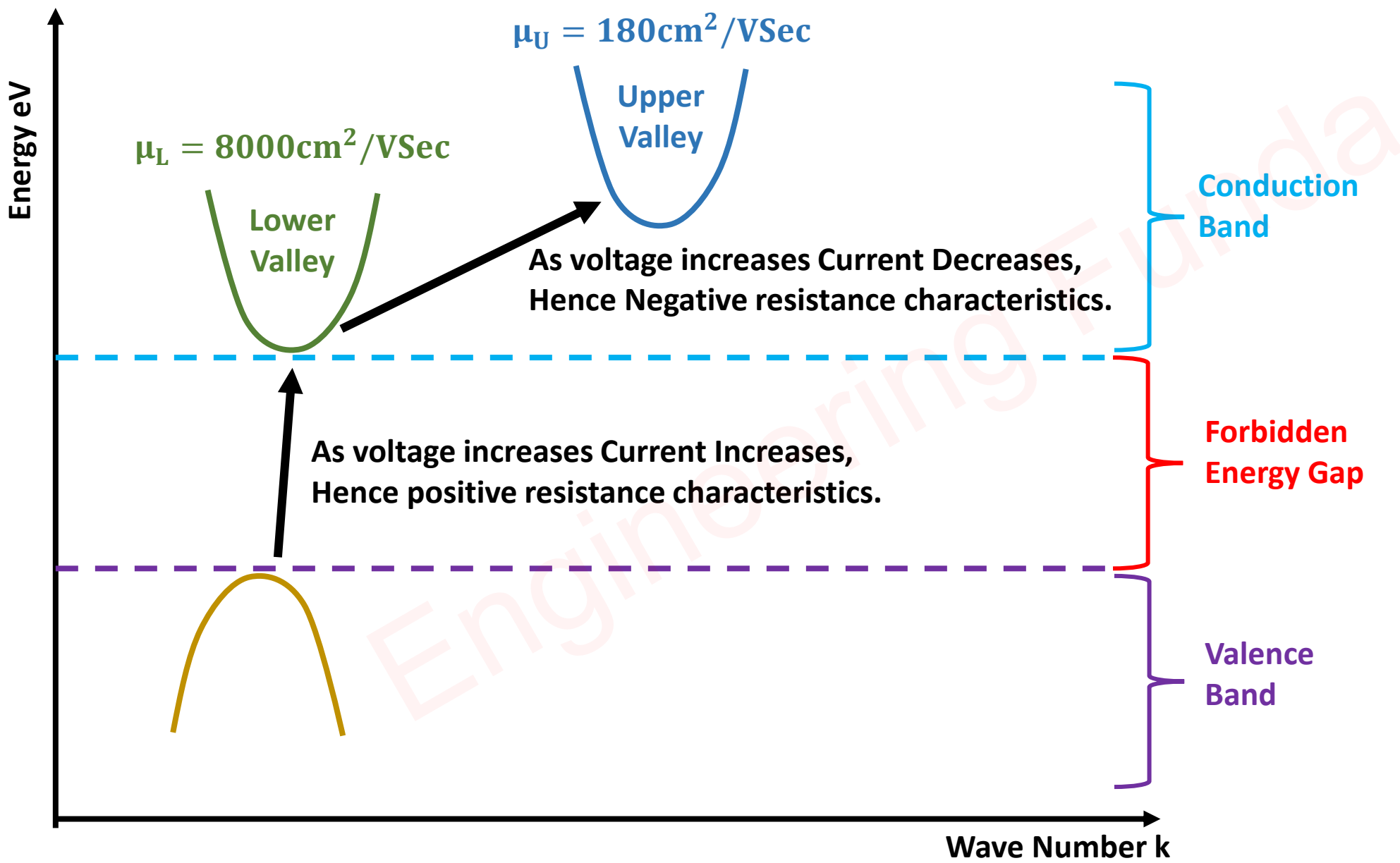
- ❑ GaAs is a poor conductor, It generate heat during conduction. So, a Heat Sink is required with Gunn Diode.

Equivalent Circuit of Gunn Diode

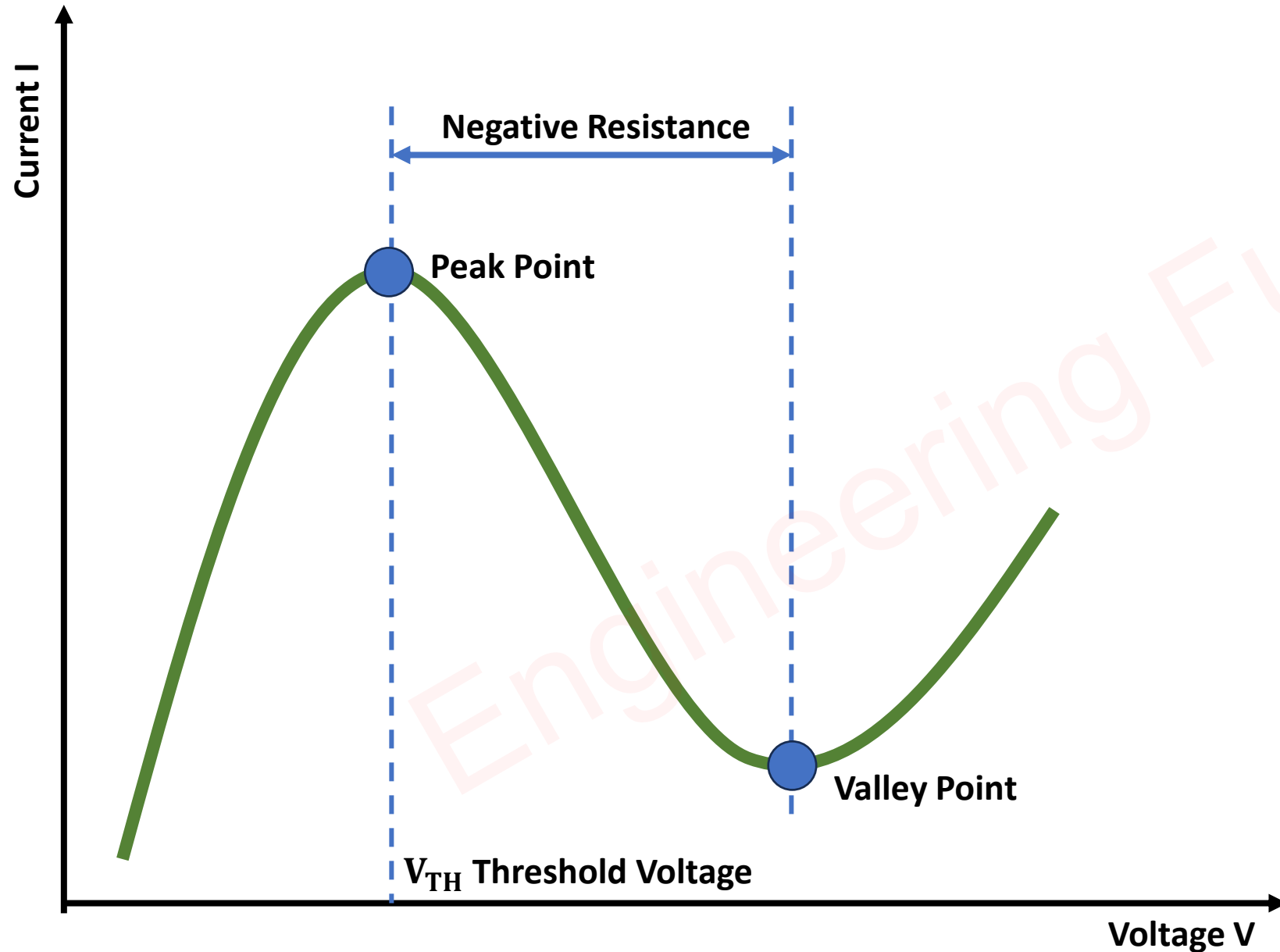


- ❑ Junction Capacitance C_j and Junction Resistance R_j changes based on Gunn Diode.
- ❑ Cap Capacitance C_P and Cap Inductance L_P is present due to metallic Cap.
- ❑ Ohmic Resistance R_S is due to ohmic contact.

Transferred Electron Effect or Gunn Effect



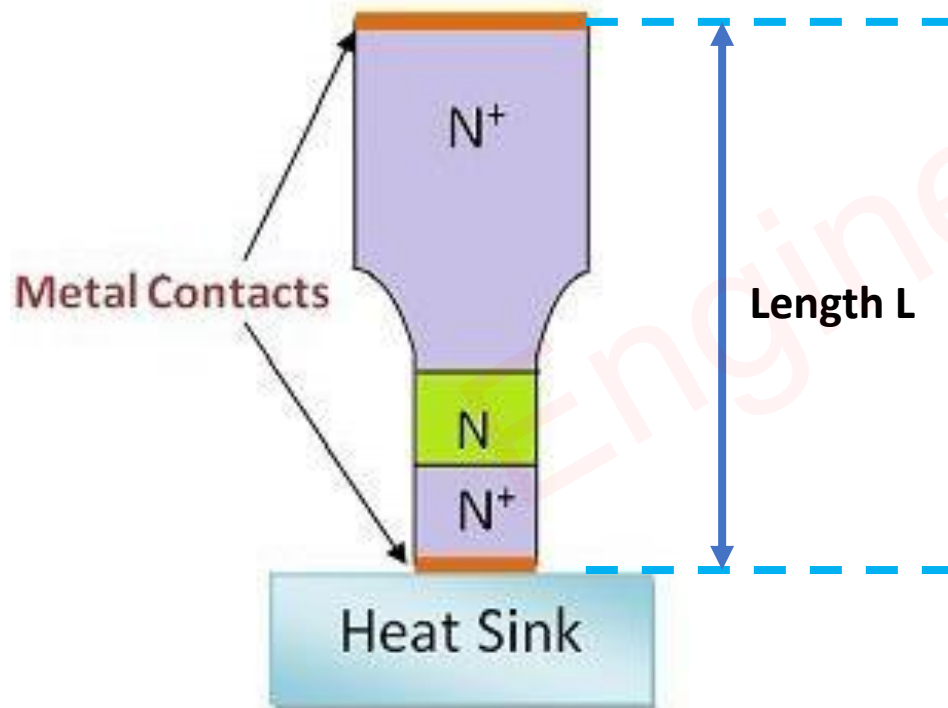
Characteristics of Gunn Diode



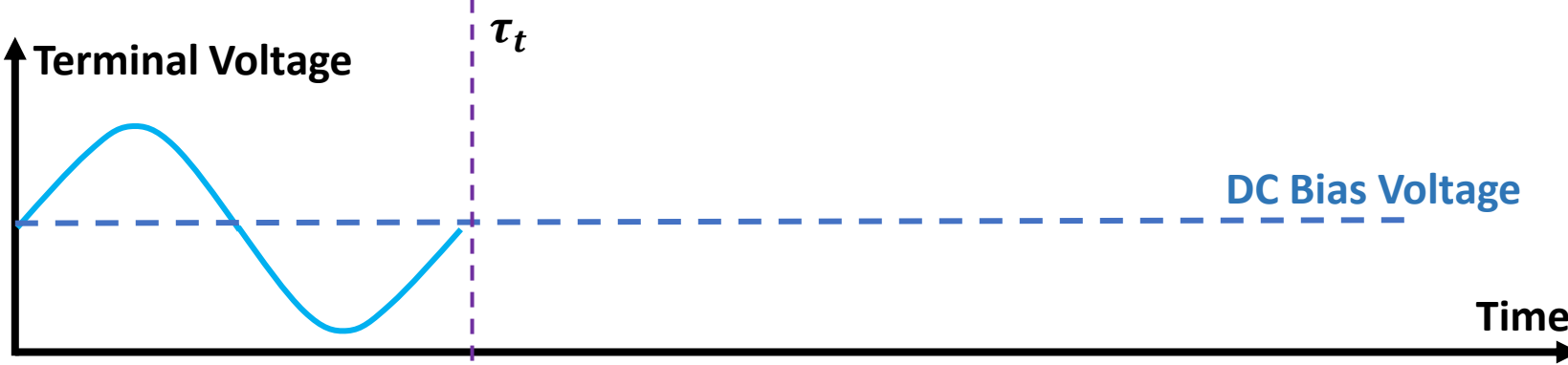
- ❑ At a Low Electric Field, Electrons are moving in the lower valley in the conduction band which has higher mobility. Hence it offers positive resistance characteristics.
- ❑ At a High Electric Field, Electrons are moving in the higher valley from the lower valley in the conduction band which has lower mobility. Hence, it offers negative resistance characteristics.

Modes of Gunn Diode

- ❑ Gunn Diode Functions in Four different modes based on operating frequency F and Length of Material L .
- ❑ Those Modes are named as:
 - Transit Time Mode
 - Delay Domain Mode
 - Quenched Domain Mode
 - Limited Space Charge Accumulation Mode

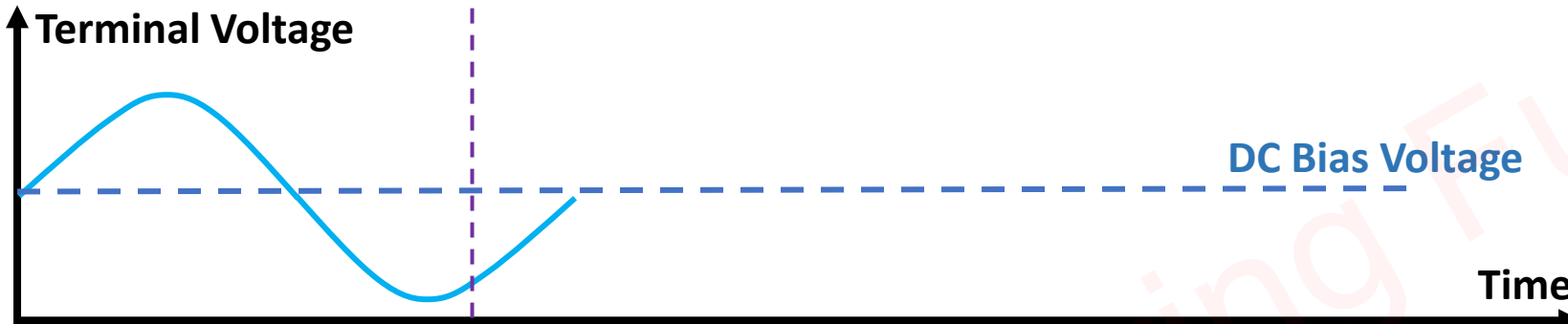


- ❖ Velocity of Charge Carriers {Electrons} $V_0 = F \times L$
- ❖ Based on V_0 , Oscillation duration τ_0 is defined.
- ❖ And based on τ_0 , Different Modes are defined.



Transit Time Mode

- Here, $v_0 = F \times L = 10^7 \text{ cm/sec}$
- Here, $\tau_0 = \tau_t$



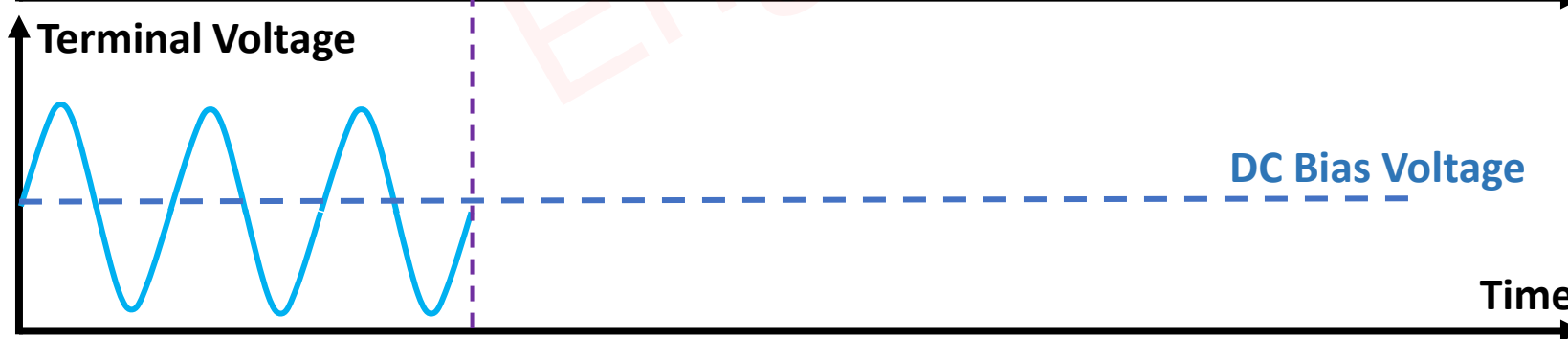
Delay Domain Mode

- Here, $v_0 = F \times L < 10^7 \text{ cm/sec}$
- Here, $\tau_0 > \tau_t$



Quenched Domain Mode

- Here, $v_0 = F \times L = 2v_d = 2 \times 10^7 \text{ cm/sec}$
- Here, $\tau_0 < \tau_t$



LSA Mode

- Here, $v_0 = F \times L > 2v_d = 2 \times 10^7 \text{ cm/sec}$
- Here, $3\tau_0 = \tau_t$

Applications of Gunn Diode

- ❑ Microwave Oscillators
- ❑ In Parametric Amplifier as Pump Source
- ❑ Used as Sensors in Microwave Systems
- ❑ In Frequency Mixers

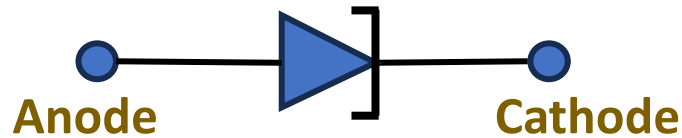
Engineering Funda

Gunn Vs IMPATT Vs TRAPPAT Vs BARITT

Parameters	Gunn	IMPATT	TRAPATT	BARITT
Operating Principle	Gunn Effect	Avalanche	Trapped Plasma	Thermionic {Punch Through}
Operating Freq.	1GHz to 100GHz	0.5GHz to 100GHz	1GHz to 10GHz	4GHz to 8GHz
Bandwidth	2%	10%	Around 5%	<2%
Output Power	Few Watts in CW Mode and 100-200W in Pulsed Mode	1W in CW Mode and 400W in Pulsed Mode	Several 100W in Pulsed Mode	Low (mW to few Watts)
Efficiency	20% in Gunn Mode	3% in CW Mode & 60% in Pulsed Mode	20-60% in Pulsed Mode	Low (2%)
Noise Figure	---	30dB	60dB	Low (<15dB)
Materials	GaAs, InP, InSb, ZnSe	Si, Ge, GaAs, InP, SiC	Si, GaAs	Si, Matel, GaAs
Harmonics	---	Less	Strong	Less
Ruggedness	Yes	Yes	Yes	Yes
Applications	Oscillator, Source, Amplifiers	Oscillator, Amplifier	Amplifier, Oscillator	Local Oscillator

Tunnel Diode/ Esaki Diode

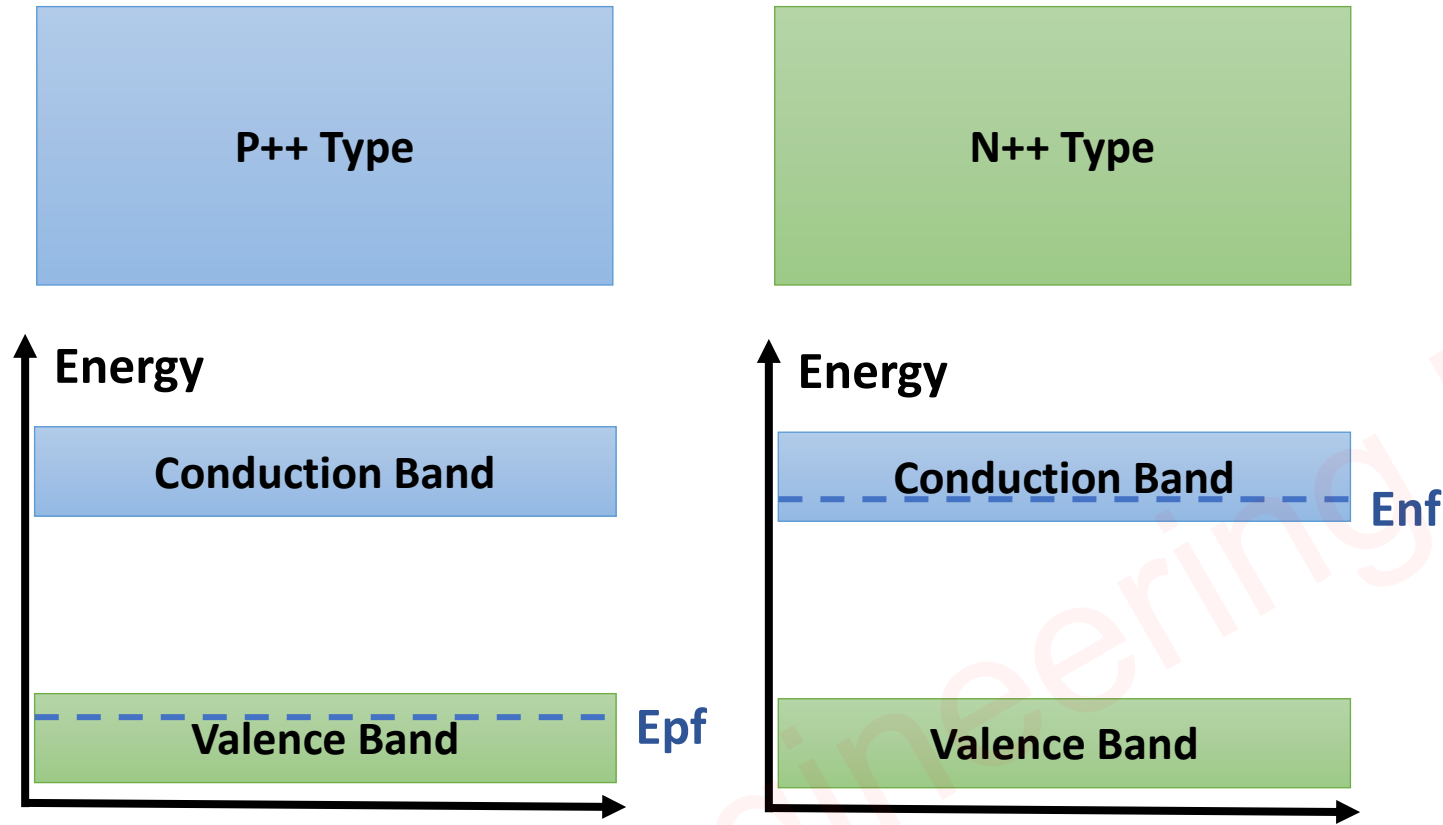
Symbol of Tunnel Diode



Basics of Tunnel Diode

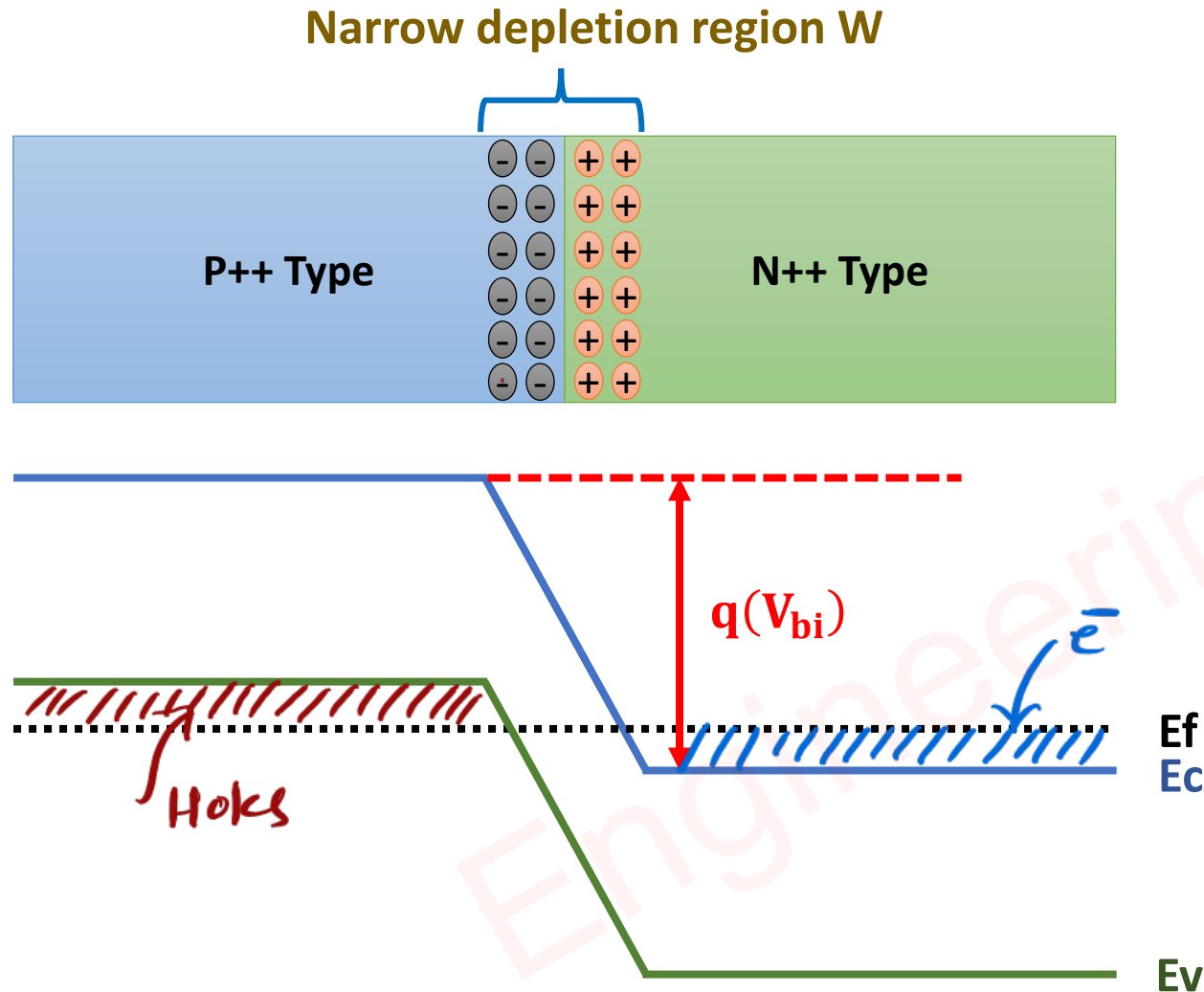
- It was invented by Leo Esaki in 1957 and he received Nobel in 1973.
- It is a heavily doped PN diode, which exhibits fast switching characteristics.
- Higher doping in semiconductors results in narrow depletion width at the PN junction of the tunnel diode.
- It is used for high-frequency applications {Oscillators & Amplifiers}.
- It works at low power.
- It exhibits negative resistance characteristics.

Structure & Energy Band of semiconductors in Tunnel Diode



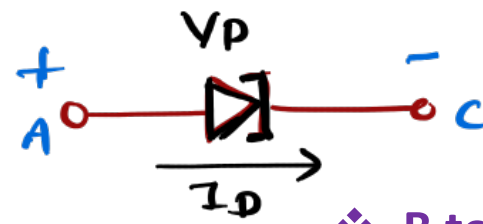
- Fermi Energy level of P++ is within the valence band and Fermi Energy level of N++ is within the conduction band. It is due to extreme doping concentration in semiconductors.

Structure & Energy Band of Tunnel Diode

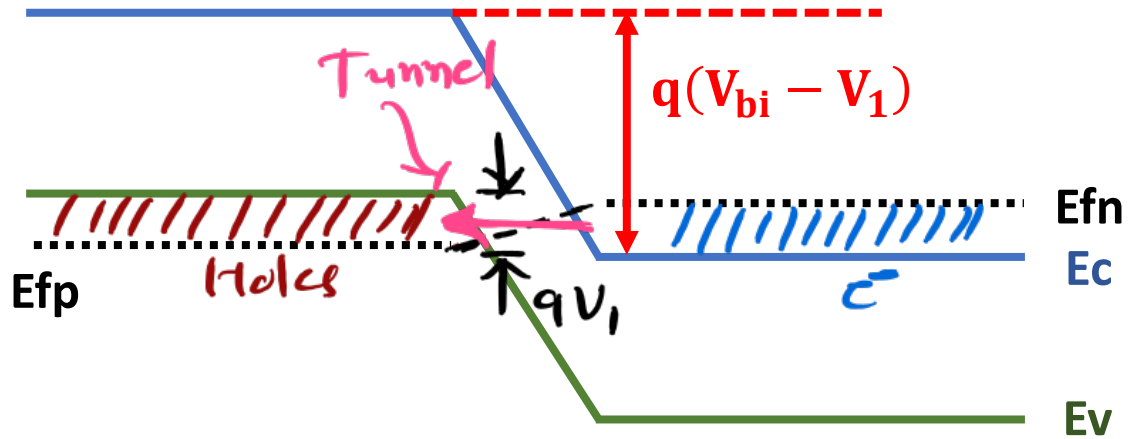


- Depletion width is very less due to higher doping concentration in semiconductors.
- In the given diagram, zero bias voltage is applied.

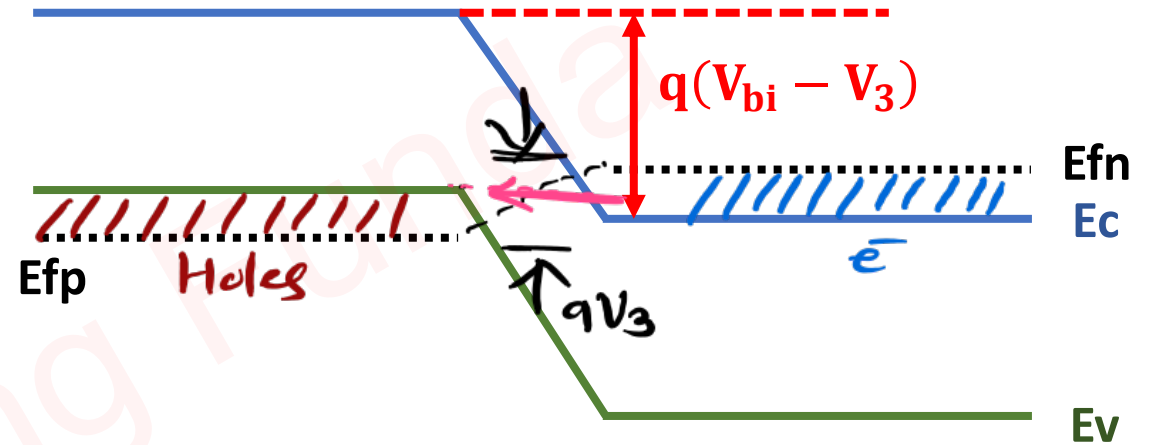
Working of Tunnel Diode



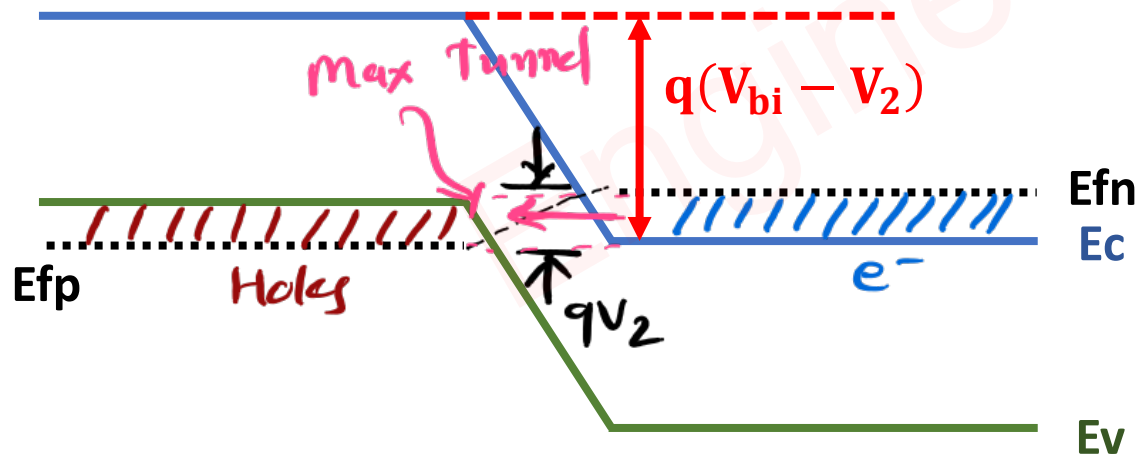
❖ 0 to A



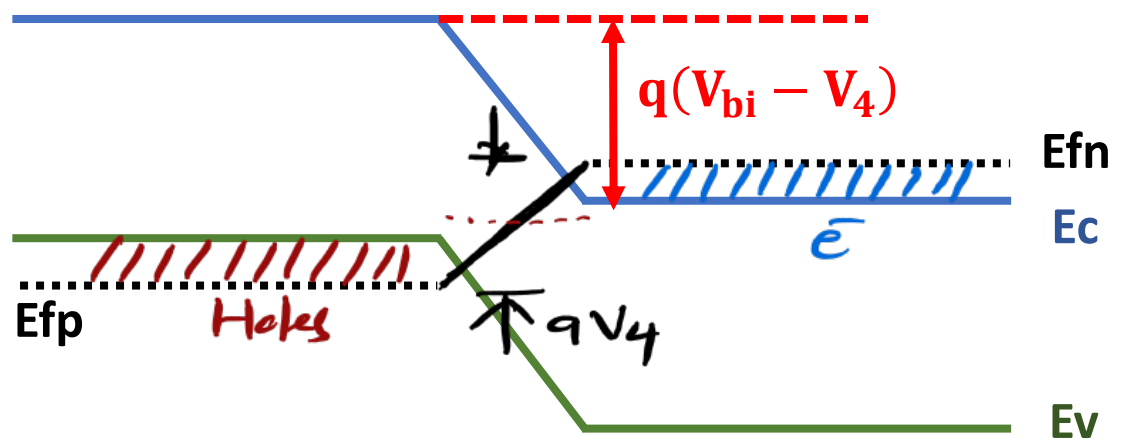
❖ **B to C**



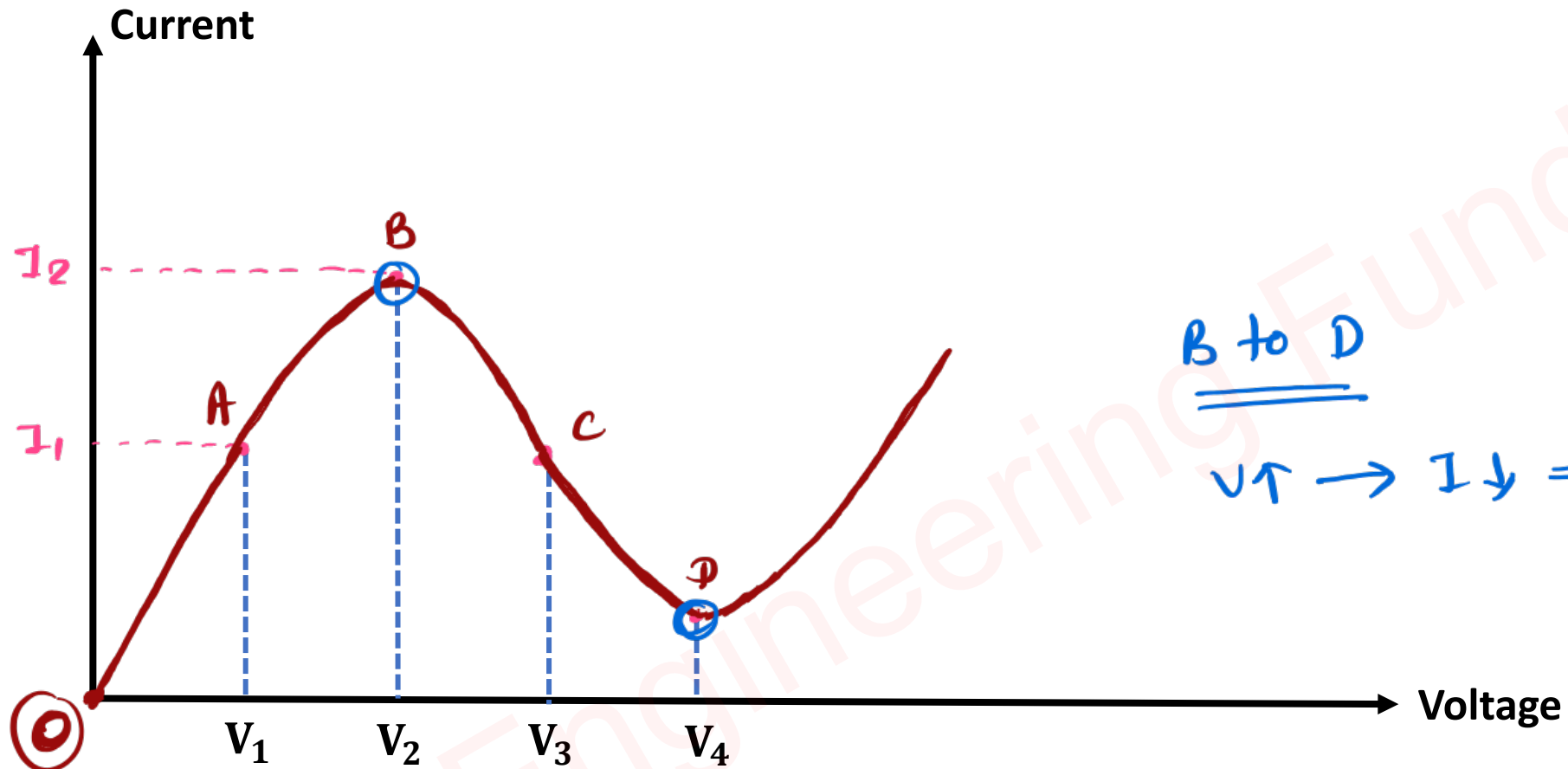
 **A to B**



❖ C to D



Characteristics of Tunnel Diode



Advantages of Tunnel Diode

- Long Life
- High-Speed Operation
- Low Noise
- Low Power Consumption

Disadvantages of Tunnel Diode

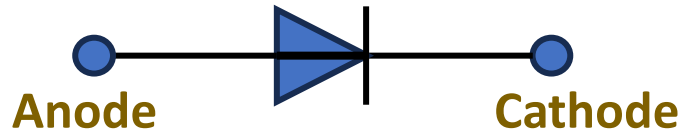
- Tunnel Diode can not be fabricated in large numbers
- Input and Output are not isolated from each other
- Low output voltage swing

Applications of Tunnel Diode

- It is used in RF and Microwave circuits.
- It is used as a Switch, Amplifier and oscillator.

PIN Diode

Symbol of PIN Diode



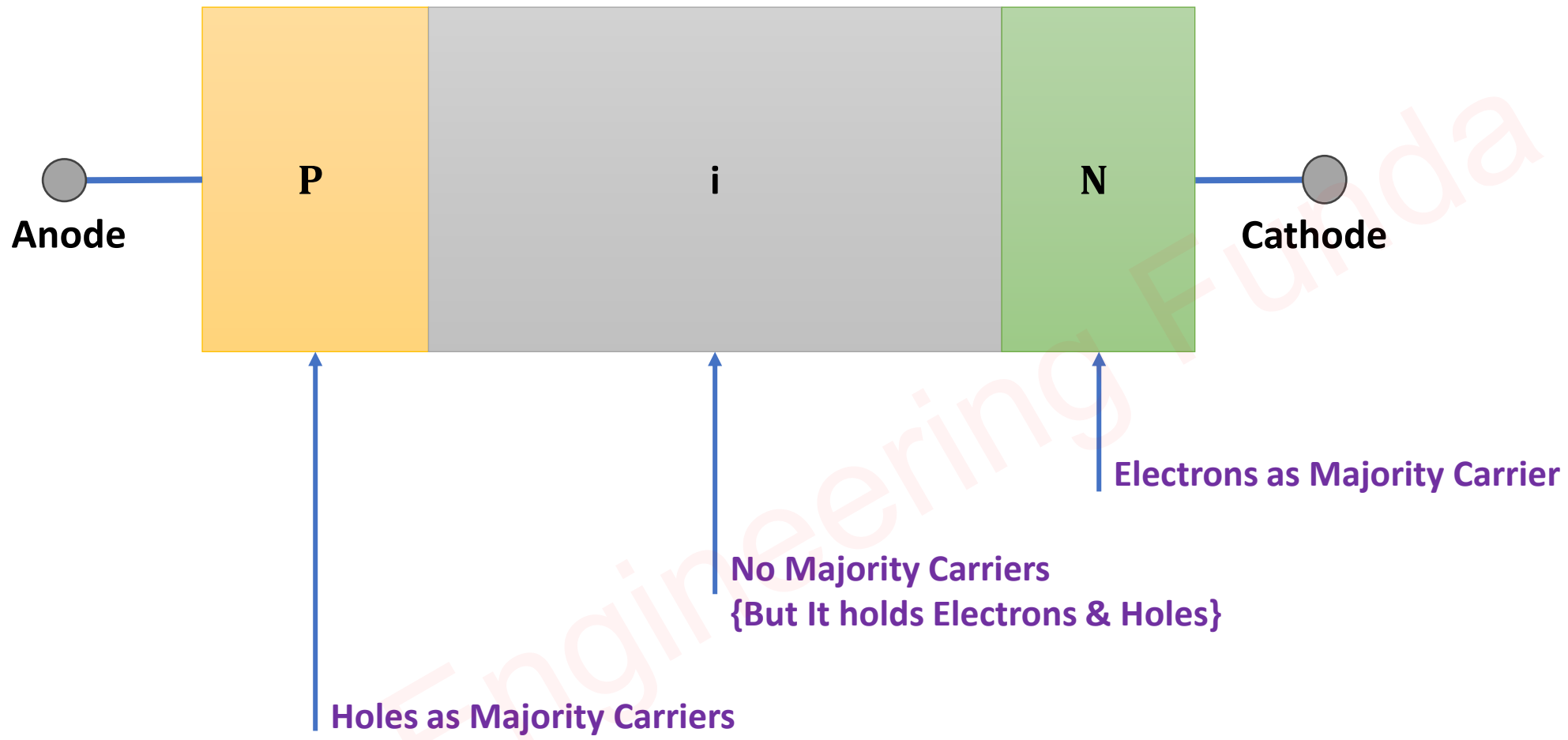
Basics of PIN Diode

- ❑ PIN Diode provides Attenuation at RF and Microwave frequencies due to the Intrinsic Layer.
- ❑ The PIN Diode can be used as an Attenuator, Leveler, and Amplitude Modulator by controlling the Forward Bias Current at RF and Microwave Frequencies.
- ❑ It has Lower Capacitance due to the Intrinsic Layer between the P-Type and N-Type Layers.

$$C = \frac{\epsilon_0 \epsilon_r A}{W}$$

- ❑ It has high Reverse Breakdown Voltage and high PIV.

Structure of PIN Diode



Characteristics of PIN Diode

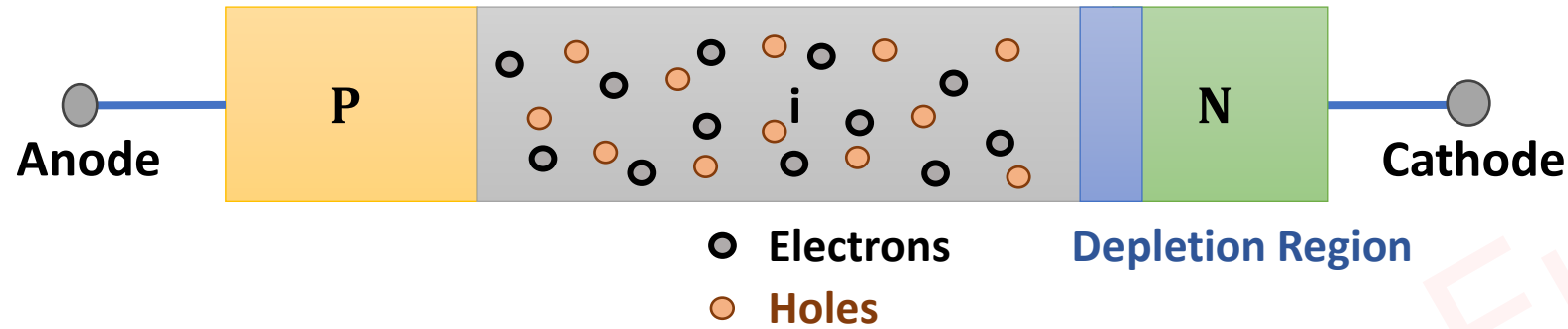
- ❑ **Low Capacitance:** Due to i-Layer {Intrinsic Layer}, it offers low capacitance across diode.

$$C = \frac{\epsilon_0 \epsilon_r A}{W}$$

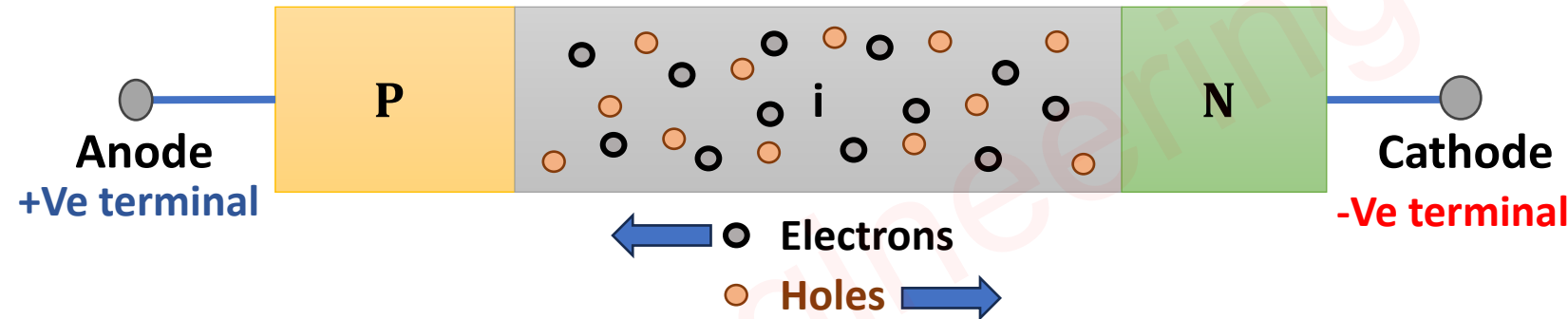
- ❑ **High-Frequency Performance:** The wide intrinsic region reduces the junction capacitance, making the PIN diode suitable for high-frequency applications.
- ❑ **High-Power Handling:** The intrinsic layer can dissipate more heat, allowing the diode to handle higher power levels.
- ❑ **Variable Resistance:** The resistance of the PIN diode can be varied by changing the bias current, which is useful in RF switching and attenuating applications.
- ❑ **High Breakdown Voltage:** Due to the Intrinsic Layer, a wider depletion region offers high breakdown voltage.
- ❑ **Carrier Storage:** a wider Intrinsic Layer can hold charge carriers, which is not happening in a normal PN junction diode.

Biasing of PIN Diode

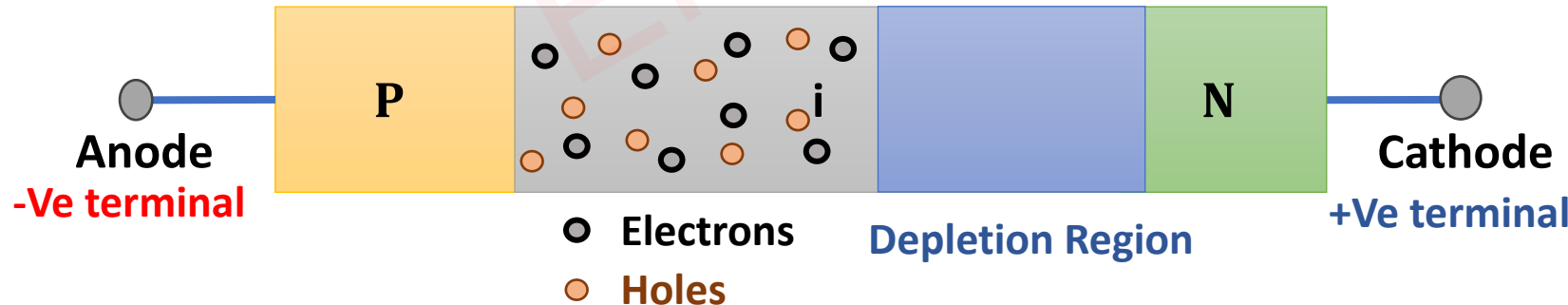
No Bias Condition



Forward Bias Condition



Reverse Bias Condition

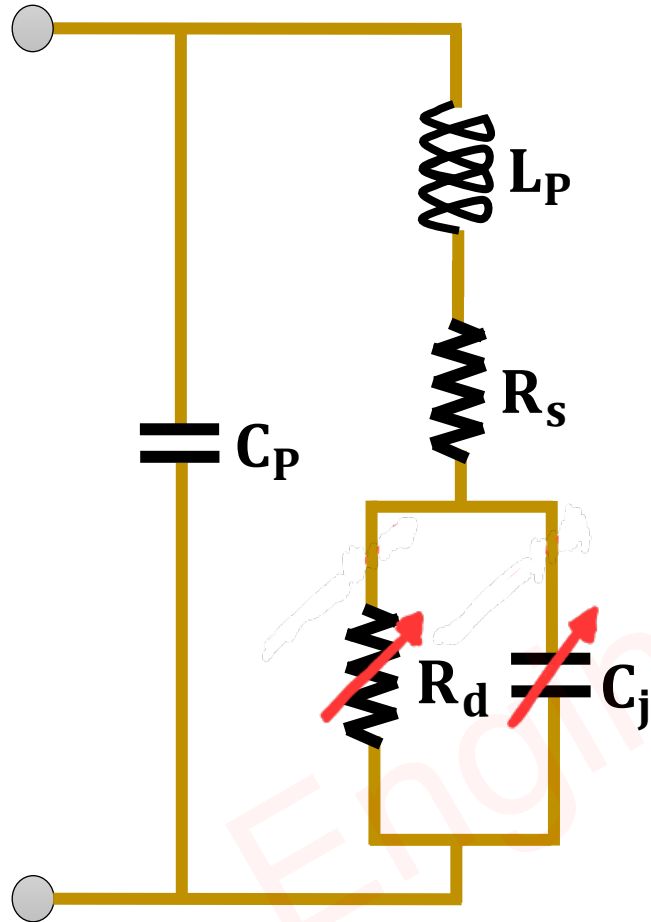


- ❑ Intrinsic Layer Holds charge carrier {Electrons and Holes}.
- ❑ In forward Bias, PIN Diode works as a variable resistor.
- ❑ Higher the forward voltage lower the forward resistance.
- ❑ In Reverse Bias, as reverse bias voltage increases, depletion region width will increase, at some voltage {Swept-out voltage}, all charges from depletion region get swept out.
- ❑ PIN Diode acts as a variable capacitor in reverse bias.

$$C = \frac{\epsilon_0 \epsilon_r A}{W}$$

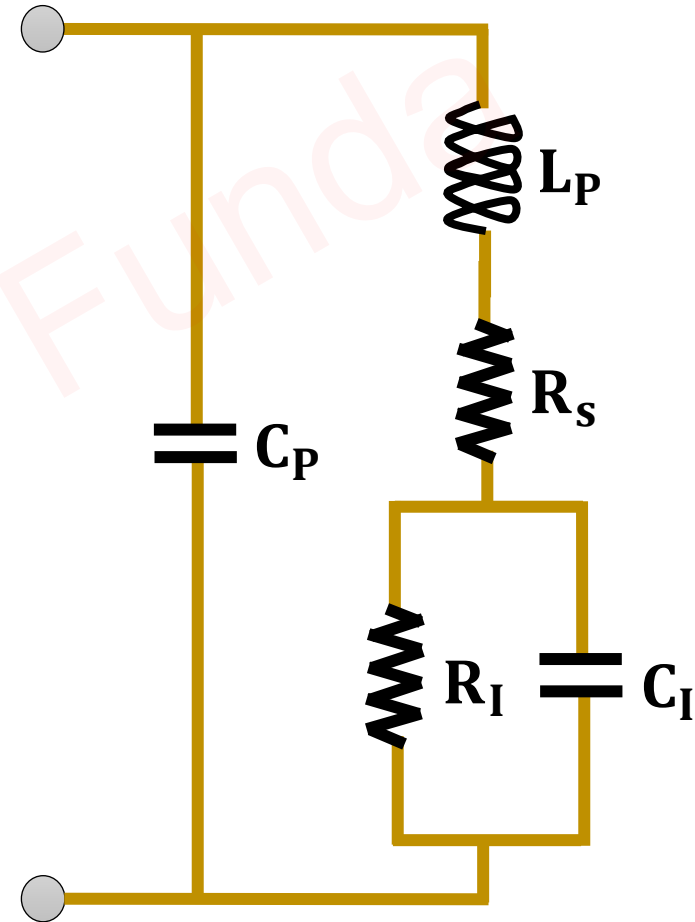
Equivalent Models of PIN Diode

Low-Frequency Model



- ❑ R_d is variable resistance in forward bias.
- ❑ C_j is variable capacitance in reverse bias.

High-Frequency Model



- ❑ R_I and C_I are RF biased components

Applications of PIN Diode

- ❑ Microwave/RF Attenuator
- ❑ PIN Diode as Switch
- ❑ RF Modulator
- ❑ Photodetector

Engineering Funda